Conformal safe sets for time-dependent vector fields:
the order-1 Wasserstein score for diffusion models and linear flow matching (proof of concept)

Companion note to *Pipeline-Agnostic Meta-Learning: an Optimal Transport Approach*

Abstract

The manuscript constrains a new model to the order-1 Wasserstein conformal region built from historical model weights. This note extends the construction from weight vectors to the time-dependent vector fields that drive diffusion and flow-matching models, so that a newly trained field is kept close to historical *safe* fields. The closed form of the order-1 score holds on any pseudometric space (Lemma 1), and split-conformal coverage holds for any cost (Proposition 1); the open question is which cost encodes safety. Viewing samplers as Gaussian-increment chains makes the path divergence between two samplers an explicit time-weighted integral of the squared field difference (Lemma 2). With this cost, every field of the region inherits the safety of the historical mixture, in total variation (Theorem 1) and, for the reverse divergence, in the sense that events that are rare under the safe samplers stay rare (Theorem 2). The direction matters: a forward cost charges at most $\log(1/(1-m))$ for an unsafe mode of mass m, the reverse cost about $m\log(m/q)$, where q is the safe probability of the unsafe region (Proposition 2). For a flow-matching velocity that is linear in its parameters, the field distance is a Mahalanobis distance whose Gram matrix is the average of the historical least-squares Gram matrices; after whitening the region is the Euclidean region of the manuscript and constrained training is a second-order cone program (Propositions 4–5). A two-dimensional toy experiment checks the coverage, the forward/reverse gap, the certificates on 200 random splits, and constrained few-shot fine-tuning against least squares and shrinkage. For prompt-conditioned models, where safe prompts induce safe fields, time collapses for straight flows (the field is constant along trajectories), the distance between noise-to-output maps carries a Wasserstein certificate without Grönwall factor, and forcing a prompt into the safe set has closed forms. A catalogue of design choices and an experiment plan close the note.

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1 Introduction

In the manuscript, a deterministic pipeline f maps a task to a weight vector $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^d$. The historical weights $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_M$ define the order-1 nonconformity score

$$s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) = \frac{1}{M(M+1)} \sum_{i=1}^{M} d(\boldsymbol{\theta}, \boldsymbol{\theta}_i) = W_{d,1}\Big(\hat{\mathbb{F}}_M, \tfrac{1}{M+1}\big(M\hat{\mathbb{F}}_M + \delta_{\boldsymbol{\theta}}\big)\Big), \tag{1}$$

and split conformal prediction turns a calibration set into a radius ε such that the region $\{\boldsymbol{\theta} : s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) \leq \varepsilon\}$ contains the pipeline's output for a new task with probability at least $1 - \beta$. Training on a new task is then restricted to this region (problem `OT1-TRAIN` in the manuscript).

Diffusion models (Ho et al. 2020, Song et al. 2021b) and flow-matching models (Lipman et al. 2023, Liu et al. 2023, Albergo and Vanden-Eijnden 2023) are not described by a weight vector at sampling time but by a *time-dependent vector field*: a score field $s(\mathbf{x}, t)$ or a velocity field $v(\mathbf{x}, t)$, integrated by a sampler from noise to data. Diffusion policies (Chi et al. 2023) use the same object to generate control actions. The question addressed here is whether the order-1 region can be built on the space of such fields, so that a field trained on a new task (possibly with few or corrupted samples) is forced to stay close to historical fields that are known to be safe.

What carries over and what has to be chosen. Lemma 1 shows that the closed form (1) only uses the triangle inequality, so it holds for any pseudometric on fields. Proposition 1 recalls that split-conformal coverage does not even need a metric: any cost that is fixed once the training fields are fixed gives a valid region. The substantive choice is therefore the cost $c(v, v_i)$ between a candidate field and a historical field. Section 4 shows that the samplers used in practice are Gaussian-increment Markov chains, for which the Kullback–Leibler divergence between the path laws of two samplers is an explicit, time-weighted integral of the squared field difference (Lemma 2). This answers the question of how to handle time: the weight given to each time step is dictated by the sampler and is not a free choice.

Safety semantics. Section 5 shows what membership in the region implies. With a cost that upper-bounds the total variation, every field in the region generates trajectories whose law is within an explicit total-variation distance of the mixture of the historical samplers (Theorem 1). With the *reverse* Kullback–Leibler divergence, measured along the candidate's own trajectories, the guarantee is of a different nature: any event that is rare under the historical safe samplers remains rare (Theorem 2). Proposition 2 explains why the direction matters, and the experiments confirm it: forward costs, which are convex and cheap, let through fields that place a few percent of their mass in an unsafe region; reverse costs do not. None of these results requires the unsafe event to be known: they hold for every event simultaneously.

Linear flow matching. Section 6 treats velocity fields that are linear in their parameters, $v_{\boldsymbol{\theta}}(\mathbf{x}, t) = \Phi(\mathbf{x}, t)\boldsymbol{\theta}$. Historical fits are least squares. The L^2 distance between two fields is a Mahalanobis distance on the parameters, and its Gram matrix under the pooled interpolation path is the average of the historical least-squares Gram matrices (Proposition 4). After whitening, the region is the Euclidean order-1 region of the manuscript, so the existing machinery (geometric median, task reweighting, gauge parametrisation) applies unchanged and `OT1-TRAIN` becomes a second-order cone program (Proposition 5). The same linear model can be sampled with a stochastic differential equation that has the same marginals (Lemma 3); the path divergence is then a quadratic form in the parameters with a Gram matrix measured along the candidate's own trajectories (Corollary 1).

Prompt-conditioned models. Section 8 specialises the construction to a trained flow-matching model conditioned on prompt embeddings, where a collection of safe prompts defines a collection of safe fields. For a straight flow the velocity is constant along each trajectory, so the time-dependent field is summarised by its slice at $t = 0$ (Lemma 7); flows trained with the linear interpolant are not straight in general (Remark 5). Comparing the noise-to-output maps of two prompts under shared noise gives a Wasserstein certificate without the Grönwall factor of Proposition 3 (Theorem 3), and forcing a prompt into the safe set has closed forms: one-step blending (Proposition 8) and, for a model affine in the prompt, a projection of the prompt embedding (Proposition 9). Section 9 catalogues the costs, regions, enforcement mechanisms and calibration procedures that the analysis supports, and Section 10 proposes experiments to test them.

Experiments. Section 7 reports a two-dimensional proof of concept: tasks are rotated and rescaled four-blob rings, and the unsafe set is a disk at the centre that no task visits. Coverage is within sampling error of the nominal level for every cost. The reverse costs exclude contaminated fields that the forward costs admit. The certificates hold on all 200 random splits, and the reverse-KL certificate is about nine times tighter than the total-variation one with exact scores, and four times with linear fits. Constrained few-shot fine-tuning improves markedly on least squares, but in this concentrated task family it is beaten by strong shrinkage towards the centre of the region. Section 11 lists the limitations.

2 Setting and notation

Tasks, pipelines and fields. Tasks τ are drawn from a meta-distribution $\mathbb{Q}$. A pipeline f maps a task to an element $v = f(\tau)$ of a set $\mathcal{V}$ of *fields*. In this note a field is a time-dependent vector field $v : \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$ (flow matching) or a score field $s : \mathbb{R}^n \times (0,T] \to \mathbb{R}^n$ (diffusion); when fields are parametrised, $\mathcal{V}$ can be identified with the parameter space. The pipeline may use internal randomness (finite data, Monte Carlo); it is applied independently to each task. We write $\mathcal{D} = (v_1, \dots, v_M)$ for the *training* (historical) fields, $v_{M+1}, \dots, v_{M+K}$ for the *calibration* fields and v_{M+K+1} for the field of a new task.

Order-1 score with an arbitrary cost.

Definition 1 (Order-1 score)**.** *Let $c : \mathcal{V} \times \mathcal{V} \to [0, +\infty]$ be measurable (not necessarily symmetric, not necessarily satisfying the triangle inequality), let $\mathcal{D} = (v_1, \dots, v_M)$ and let w be a probability vector on $[\![M]\!]$. The weighted order-1 score of $v \in \mathcal{V}$ is*

$$S_c^w(v; \mathcal{D}) \coloneqq \frac{1}{M+1} \sum_{i=1}^{M} w_i \, c(v, v_i), \qquad S_c(v; \mathcal{D}) \coloneqq S_c^{(1/M, \dots, 1/M)}(v; \mathcal{D}) = \frac{1}{M(M+1)} \sum_{i=1}^{M} c(v, v_i). \tag{2}$$

For $c = d$ a metric on weights, S_c is the score $s_{d,1}$ of (1) and S_c^w is the task-adaptive score $g_{d,1}$ of the manuscript. Note that

$$S_c^w(v; \mathcal{D}) \le \varepsilon \quad \Longleftrightarrow \quad \sum_{i=1}^{M} w_i \, c(v, v_i) \le (M+1)\, \varepsilon. \tag{3}$$

The quantity $(M+1)\varepsilon$ is therefore the budget on the w-average cost, and it is the quantity that appears in all safety statements below.

Definition 2 (Split-conformal threshold and region)**.** *Let $\beta \in (0,1)$, $k \coloneqq \lceil (K+1)(1-\beta) \rceil$, and let $S_j \coloneqq S_c(v_{M+j}; \mathcal{D})$, $j \in [\![K]\!]$, be the calibration scores. The threshold ε is the k-th smallest of $S_1, \dots, S_K$ if $k \le K$ and $+\infty$ otherwise; this is $C_{1-\beta}(S_1, \dots, S_K)$ in the notation of the manuscript. The region is $\bar{\mathcal{V}} \coloneqq \{v \in \mathcal{V} : S_c(v; \mathcal{D}) \le \varepsilon\}$.*

Probability metrics. For probability measures P, Q on a common measurable space, $\mathrm{TV}(P,Q) \coloneqq \sup_A |P(A) - Q(A)|$ (supremum over measurable sets), $\mathrm{KL}(P\|Q) \coloneqq \mathbb{E}_P[\log \frac{dP}{dQ}]$ if $P \ll Q$ and $+\infty$ otherwise, and for $p, q \in [0,1]$ the binary divergence is $\mathrm{kl}(p\|q) \coloneqq p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}$ with $0 \log 0 = 0$.

3 The order-1 score on a space of fields

Lemma 1 (Closed form on a pseudometric space)**.** *Let $(\mathcal{V}, d)$ be a pseudometric space: $d \ge 0$ is symmetric, satisfies the triangle inequality and $d(v,v) = 0$ (distinct points may be at distance zero). Let $v_1, \dots, v_M, v \in \mathcal{V}$, let w be a probability vector on $[\![M]\!]$, and set*

$$\mu_w \coloneqq \sum_{i=1}^{M} w_i \, \delta_{v_i}, \qquad \nu_w \coloneqq \frac{M}{M+1} \, \mu_w + \frac{1}{M+1} \, \delta_v.$$

Let $W_{d,1}(\mu, \nu) \coloneqq \min_{\pi \in \Pi(\mu,\nu)} \int d \, d\pi$, where $\Pi(\mu,\nu)$ is the set of couplings of the two finitely supported measures. Then

$$W_{d,1}(\mu_w, \nu_w) = \frac{1}{M+1} \sum_{i=1}^{M} w_i \, d(v_i, v) = S_d^w(v; \mathcal{D}).$$

In particular, for $w_i = 1/M$, $W_{d,1}(\hat{\mathbb{F}}_M, \frac{1}{M+1}(M\hat{\mathbb{F}}_M + \delta_v)) = \frac{1}{M(M+1)} \sum_i d(v_i, v)$.

Proof: Step 1 (a feasible plan: the upper bound). Let $\pi^\star \coloneqq \sum_i \frac{M w_i}{M+1} \delta_{(v_i, v_i)} + \sum_i \frac{w_i}{M+1} \delta_{(v_i, v)}$. Its first marginal is $\sum_i w_i \delta_{v_i} = \mu_w$ and its second marginal is $\frac{M}{M+1}\mu_w + \frac{1}{M+1}\delta_v = \nu_w$, since $\sum_i w_i = 1$. Hence $\pi^\star \in \Pi(\mu_w, \nu_w)$ and, using $d(v_i, v_i) = 0$, $W_{d,1}(\mu_w, \nu_w) \le \int d \, d\pi^\star = \frac{1}{M+1} \sum_i w_i \, d(v_i, v)$.

Step 2 (every plan costs at least as much: the lower bound). Let $\pi \in \Pi(\mu_w, \nu_w)$ and let $g(x) \coloneqq d(x, v)$. The triangle inequality $d(x,v) \le d(x,y) + d(y,v)$ gives $d(x,y) \ge g(x) - g(y)$ for all $x, y \in \mathcal{V}$. Integrating against π and using its marginals,

$$\int d \, d\pi \;\ge\; \int g \, d\mu_w - \int g \, d\nu_w = \sum_i w_i \, d(v_i, v) - \frac{M}{M+1} \sum_i w_i \, d(v_i, v) - \frac{1}{M+1} \, d(v,v) = \frac{1}{M+1} \sum_i w_i \, d(v_i, v).$$

Step 3. Steps 1 and 2 give the equality. □

Remark 1. *Lemma 1 is `lem:trivial_W1` of the manuscript (case $w_i = 1/M$, d a metric on $\mathbb{R}^d$), and its weighted form is equation `eq:adaptive_W1`. Step 2 replaces the "intermediate destination" argument of the manuscript by a single application of the triangle inequality, which is why the result holds on any pseudometric space, in particular on spaces of time-dependent fields with the distances of Section 6.*

Proposition 1 (Coverage for an arbitrary cost)**.** *Let c be a cost as in Definition 1, possibly depending on an auxiliary random element Ξ (for instance the noise of Monte Carlo probe trajectories), and assume that, conditionally on $(\mathcal{D}, \Xi)$, the vector $(v_{M+1}, \dots, v_{M+K}, v_{M+K+1})$ is exchangeable. Let $S_j := S_c(v_{M+j}; \mathcal{D})$ for $j \in [\![K+1]\!]$, and let ε be the threshold of Definition 2. Then*

$$\mathbb{P}\big(S_c(v_{M+K+1}; \mathcal{D}) \le \varepsilon \,\big|\, \mathcal{D}, \Xi\big) \ge \frac{k}{K+1} \ge 1 - \beta, \qquad k = \lceil (K+1)(1-\beta) \rceil.$$

Proof: Step 1 (trivial case). If $k > K$, then $\varepsilon = +\infty$ and the event has probability one.

Step 2 (event inclusion). Let $k \le K$ and let $s_{[1]} \le \cdots \le s_{[K+1]}$ be the order statistics of all $K+1$ scores $S_1, \dots, S_{K+1}$. For $j \in [\![K+1]\!]$ let $E_j := \{S_j > s_{[k]}\}$. If $S_{K+1} > \varepsilon$, then the k smallest calibration scores are all strictly smaller than S_{K+1}; these are k of the $K+1$ scores, so $s_{[k]} \le \varepsilon < S_{K+1}$, i.e. E_{K+1} holds. Hence $\{S_{K+1} > \varepsilon\} \subseteq E_{K+1}$.

Step 3 (deterministic count). At least k indices j satisfy $S_j \le s_{[k]}$, so at most $K+1-k$ indices satisfy E_j: $\sum_{j=1}^{K+1} \mathbb{1}_{E_j} \le K+1-k$ pointwise.

Step 4 (exchangeability). Conditionally on $(\mathcal{D}, \Xi)$ the score function $S_c(\cdot; \mathcal{D})$ is fixed, so $(S_1, \dots, S_{K+1})$ is exchangeable and $\mathbb{P}(E_j \mid \mathcal{D}, \Xi)$ does not depend on j. Taking the conditional expectation of Step 3, $(K+1)\,\mathbb{P}(E_{K+1} \mid \mathcal{D}, \Xi) \le K+1-k$.

Step 5. By Steps 2 and 4, $\mathbb{P}(S_{K+1} \le \varepsilon \mid \mathcal{D}, \Xi) \ge 1 - \frac{K+1-k}{K+1} = \frac{k}{K+1} \ge 1 - \beta$. □

Remark 2 (What the statement does and does not say)**.** *(i) The probability is conditional on the training fields (and on Ξ) and marginal over the calibration fields and the new field. A statement conditional on the calibration fields as well requires the training-conditional (PAC) form of Vovk (2013); this is the same distinction as the one flagged for `cor:ot1_cp` of the manuscript. (ii) What is covered is the field $f(\tau)$ that the pipeline would produce on the new task (e.g. with ample data), not a field fitted on few samples; restricting few-shot training to $\bar{\mathcal{V}}$ therefore excludes the pipeline's answer with probability at most β. (iii) Exchangeability holds when tasks are i.i.d. and the pipeline is applied independently to each of them. If historical safe fields are selected by an audit, the guarantee is relative to the population of fields that pass the audit. For histories that drift over time, the weighted conformal procedure of Barber et al. (2023) applies. (iv) Monte Carlo estimates of the cost are allowed as long as their randomness is part of Ξ, i.e. drawn once and shared by all scores ("frozen probes"): the coverage is then exact for the estimated score. The safety implications of Section 5, on the other hand, are stated for the exact cost (Remark 3).*

4 Samplers as Gaussian-increment chains

Definition 3 (Gaussian-increment chain and path law)**.** *Fix $J \in \mathbb{N}$, step sizes $h_0, \dots, h_{J-1} > 0$, noise levels $\sigma_0, \dots, \sigma_{J-1} \ge 0$, an initial law μ_0 on $\mathbb{R}^n$ and measurable drifts $b_k : \mathbb{R}^n \to \mathbb{R}^n$, $k \in \{0, \dots, J-1\}$. The chain is $X_0 \sim \mu_0$ and*

$$X_{k+1} = X_k + h_k\, b_k(X_k) + \sigma_k\, \boldsymbol{\xi}_k, \qquad k = 0, \dots, J-1,$$

with $\boldsymbol{\xi}_0, \dots, \boldsymbol{\xi}_{J-1}$ i.i.d. $\mathcal{N}(0, I_n)$ and independent of X_0. Its path law is the law P^b of $(X_0, \dots, X_J)$ on $(\mathbb{R}^n)^{J+1}$. A sampler turns a field v into drifts $b[v]$; we write $P^v := P^{b[v]}$, $P^{(i)} := P^{v_i}$ for the historical fields, and $\bar{P}_w := \sum_i w_i P^{(i)}$ for their mixture ($\bar{P} := \bar{P}_{(1/M,\dots,1/M)}$). An event *is a measurable set $A \subseteq (\mathbb{R}^n)^{J+1}$; in the experiments $A = \{\|\mathbf{x}_J\| \le r_A\}$ concerns the final sample only.*

The samplers used below are the following; all of them start from $\mu_0 = \mathcal{N}(0, I_n)$.

- *Diffusion (variance-preserving, unit noise rate).* The forward process $d\mathbf{x} = -\frac{1}{2}\mathbf{x}\,dt + dB_t$ has marginals $\mathbf{x}_t = \kappa_t \mathbf{x}_0 + \varsigma_t \boldsymbol{\xi}$ with $\kappa_t = e^{-t/2}$ and $\varsigma_t^2 = 1 - \kappa_t^2$. On a decreasing grid $T = t_0 > t_1 > \cdots > t_J = t_{\min}$ with $h_k = t_k - t_{k+1}$, the generative chain of a score field s is $b[s]_k(\mathbf{y}) = \frac{1}{2}\mathbf{y} + s(\mathbf{y}, t_k)$ with $\sigma_k^2 = h_k$ (Euler–Maruyama discretisation of the reverse-time SDE of Song et al. (2021b)).
- *Flow matching, ODE sampler.* On the grid $0 = t_0 < \cdots < t_J = 1$, $b[v]_k(\mathbf{y}) = v(\mathbf{y}, t_k)$ and $\sigma_k = 0$ (explicit Euler).
- *Flow matching, SDE sampler.* For $\eta > 0$, $b[v]_k(\mathbf{y}) = (1 + \eta t_k)\, v(\mathbf{y}, t_k) - \eta \mathbf{y}$ and $\sigma_k^2 = 2\eta(1 - t_k)h_k$; Lemma 3 explains why this chain samples (approximately) the same distribution as the ODE.

Lemma 2 (Exact path divergence between two chains)**.** *Let P^b and $P^{b'}$ be two chains as in Definition 3 with the same J, the same h_k, the same $\sigma_k > 0$ and the same μ_0, and drifts b, b'. Then*

$$\mathrm{KL}(P^b \| P^{b'}) = \sum_{k=0}^{J-1} \frac{h_k^2}{2\sigma_k^2}\; \mathbb{E}_{P^b}\left\|b_k(X_k) - b'_k(X_k)\right\|^2 \;\in [0, +\infty].$$

Proof: Step 1 (Markov kernels). By Definition 3, given $(X_0, \dots, X_k)$ the next state of P^b has law $\mathcal{N}(X_k + h_k b_k(X_k), \sigma_k^2 I_n)$, and that of $P^{b'}$ has law $\mathcal{N}(X_k + h_k b'_k(X_k), \sigma_k^2 I_n)$; the conditional laws depend on the past only through X_k.

Step 2 (chain rule). By the chain rule for relative entropy (Cover and Thomas 2006, Thm. 2.5.3), $\mathrm{KL}(P^b\|P^{b'}) = \mathrm{KL}(\mu_0\|\mu_0) + \sum_{k=0}^{J-1} \mathbb{E}_{P^b}\big[\mathrm{KL}\big(P^b(\cdot \mid X_{0:k}) \,\|\, P^{b'}(\cdot \mid X_{0:k})\big)\big]$, where every term is nonnegative, so the identity holds in $[0,+\infty]$. The first term is zero.

Step 3 (Gaussian divergence). For two Gaussians with the same covariance, $\mathrm{KL}(\mathcal{N}(\mathbf{m},\sigma^2 I)\|\mathcal{N}(\mathbf{m}',\sigma^2 I)) = \|\mathbf{m}-\mathbf{m}'\|^2/(2\sigma^2)$. With the means of Step 1, $\|\mathbf{m}-\mathbf{m}'\| = h_k\|b_k(X_k) - b'_k(X_k)\|$, which gives the k-th term. □

Corollary 1 (Diffusion and flow-matching chains)**.** *(i) For the diffusion chain of two score fields s, s':*

$$\mathrm{KL}(P^s\|P^{s'}) = \frac{1}{2}\sum_{k=0}^{J-1} h_k\ \mathbb{E}_{P^s}\left\|s(X_k,t_k) - s'(X_k,t_k)\right\|^2.$$

(ii) For the flow-matching SDE chain of two velocity fields v, v':

$$\mathrm{KL}(P^v\|P^{v'}) = \frac{1}{2}\sum_{k=0}^{J-1} \omega_k\ \mathbb{E}_{P^v}\left\|v(X_k,t_k) - v'(X_k,t_k)\right\|^2, \qquad \omega_k \coloneqq \frac{h_k(1+\eta t_k)^2}{2\eta(1-t_k)}.$$

(iii) If moreover $v = \Phi\boldsymbol{\theta}$ and $v' = \Phi\boldsymbol{\theta}'$ for a feature map $\Phi : \mathbb{R}^n \times [0,1] \to \mathbb{R}^{n\times D}$, then

$$\mathrm{KL}(P^{\boldsymbol{\theta}}\|P^{\boldsymbol{\theta}'}) = \tfrac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\theta}')^\top G_{\mathrm{KL}}(\boldsymbol{\theta})\,(\boldsymbol{\theta}-\boldsymbol{\theta}'), \qquad G_{\mathrm{KL}}(\boldsymbol{\theta}) \coloneqq \sum_{k=0}^{J-1} \omega_k\ \mathbb{E}_{P^{\boldsymbol{\theta}}}\left[\Phi(X_k,t_k)^\top \Phi(X_k,t_k)\right].$$

The Gram matrix $G_{\mathrm{KL}}(\boldsymbol{\theta})$ is computed along the trajectories of the first *argument.*

Proof: (i) Lemma 2 with $b_k - b'_k = s(\cdot,t_k) - s'(\cdot,t_k)$ and $\sigma_k^2 = h_k$ gives $h_k^2/(2h_k) = h_k/2$. (ii) Here $b_k - b'_k = (1+\eta t_k)(v-v')(\cdot,t_k)$ and $\sigma_k^2 = 2\eta(1-t_k)h_k$, so the factor is $h_k^2(1+\eta t_k)^2/(4\eta(1-t_k)h_k) = \omega_k/2$. (iii) Substitute $v - v' = \Phi(\boldsymbol{\theta}-\boldsymbol{\theta}')$ in (ii) and use $\|\Phi\mathbf{a}\|^2 = \mathbf{a}^\top\Phi^\top\Phi\mathbf{a}$ and linearity of the expectation. □

Two consequences are worth stating. First, the time dependence of the fields needs no separate treatment: the divergence integrates the squared field difference over time with the weights $h_k/2$ (diffusion) or $\omega_k/2$ (flow matching), which are fixed by the sampler. The weight ω_k grows like $1/(1-t_k)$, so the late, near-data steps count most. In continuous time, (i) is the Girsanov identity used by Song et al. (2021a) and Chen et al. (2023), and the per-step sum of (i) is the regulariser towards a single pretrained model used by Fan et al. (2023, Lemma 4.2). Second, a deterministic sampler ($\sigma_k = 0$) has no such formula: two ODE samplers driven by different fields generally have mutually singular path laws. Deterministic samplers are treated through couplings in Proposition 3.

Lemma 3 (Marginal-preserving SDE for linear-interpolant flow matching)**.** *Let $\mathbf{x}_0 \sim \mathcal{N}(0, I_n)$ and $\mathbf{x}_1 \sim p$ be independent, $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$, let p_t be the law of $\mathbf{x}_t$ and $v^\star(\mathbf{x},t) \coloneqq \mathbb{E}[\mathbf{x}_1 - \mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}]$. For $t \in [0,1)$: (i) p_t has a smooth positive density and $\nabla \log p_t(\mathbf{x}) = \dfrac{t\,v^\star(\mathbf{x},t) - \mathbf{x}}{1-t}$; (ii) (formal) for every $\eta \geq 0$, the SDE $dX_t = \big[(1+\eta t)v^\star(X_t,t) - \eta X_t\big]dt + \sqrt{2\eta(1-t)}\,dB_t$, $X_0 \sim \mathcal{N}(0,I_n)$, has the marginals p_t.*

Proof: (i) *Step 1.* Given $\mathbf{x}_1$, $\mathbf{x}_t \sim \mathcal{N}(t\mathbf{x}_1, (1-t)^2 I_n)$, so $p_t(\mathbf{x}) = \mathbb{E}_{\mathbf{x}_1}[g_t(\mathbf{x} - t\mathbf{x}_1)]$ with g_t the $\mathcal{N}(0,(1-t)^2 I_n)$ density; p_t is a Gaussian convolution, hence smooth and positive. *Step 2.* Differentiating under the expectation (the derivatives of g_t are bounded) and applying Bayes' rule, $\nabla \log p_t(\mathbf{x}) = -\,\mathbb{E}[\mathbf{x} - t\mathbf{x}_1 \mid \mathbf{x}_t = \mathbf{x}]/(1-t)^2$. Since $\mathbf{x}_t - t\mathbf{x}_1 = (1-t)\mathbf{x}_0$, this equals $-\,\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}]/(1-t)$. *Step 3.* Taking conditional expectations in $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$ and in the definition of $v^\star$ gives $\mathbf{x} = (1-t)\,\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}] + t\,\mathbb{E}[\mathbf{x}_1 \mid \mathbf{x}_t = \mathbf{x}]$ and $v^\star(\mathbf{x},t) = \mathbb{E}[\mathbf{x}_1 \mid \mathbf{x}_t = \mathbf{x}] - \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}]$. Eliminating $\mathbb{E}[\mathbf{x}_1 \mid \mathbf{x}_t = \mathbf{x}]$, $\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}] = \mathbf{x} - t\,v^\star(\mathbf{x},t)$. Insert in Step 2.

(ii) The path (p_t) solves the continuity equation $\partial_t p_t + \nabla\cdot(p_t v^\star) = 0$ (Lipman et al. 2023, Liu et al. 2023, Albergo and Vanden-Eijnden 2023). Write $g^2 \coloneqq 2\eta(1-t)$. By (i), $\eta t\, v^\star - \eta\mathbf{x} = \eta(1-t)\frac{tv^\star - \mathbf{x}}{1-t} = \frac{g^2}{2}\nabla \log p_t$, so the drift is $v^\star + \frac{g^2}{2}\nabla\log p_t$. Its Fokker–Planck operator applied to p_t is $-\nabla\cdot(p_t v^\star) - \frac{g^2}{2}\nabla\cdot(p_t\nabla\log p_t) + \frac{g^2}{2}\Delta p_t = -\nabla\cdot(p_t v^\star) = \partial_t p_t$, because $p_t\nabla\log p_t = \nabla p_t$. Under uniqueness of the Fokker–Planck solution from $\mathcal{N}(0,I_n)$ (regularity is not checked here; see Albergo et al. 2025 for the rigorous general statement), the marginals coincide. □

The safety results never use Lemma 3: they concern the chains that are actually simulated, whatever field drives them. The lemma only explains why sampling a learned velocity with the SDE chain targets the same distribution as the ODE.

5 Safety transfer from the order-1 region

Lemma 4 (A concave link between KL and TV)**.** *Let $\varphi(c) \coloneqq \min\big\{c/2,\ \sqrt{1-e^{-c^2/2}}\big\}$ for $c \in [0,+\infty]$ ($\varphi(+\infty) = 1$). (i) φ is concave and nondecreasing on $[0,+\infty)$ and $\varphi \leq 1$. (ii) For all probability measures P, Q, $\mathrm{TV}(P,Q) \leq \varphi\big(\sqrt{2\mathrm{KL}(P\|Q)}\big)$ and $\mathrm{TV}(P,Q) \leq \varphi\big(\sqrt{2\mathrm{KL}(Q\|P)}\big)$.*

Proof: (ii) With $c = \sqrt{2\mathrm{KL}}$, Pinsker's inequality (Tsybakov 2009, Lemma 2.5) reads $\mathrm{TV} \le \sqrt{\mathrm{KL}/2} = c/2$, and the Bretagnolle–Huber inequality (Bretagnolle and Huber 1979, Canonne 2022) reads $\mathrm{TV} \le \sqrt{1 - e^{-\mathrm{KL}}} = \sqrt{1 - e^{-c^2/2}}$. Both apply to either direction because TV is symmetric. (i) *Step 1.* $c \mapsto c/2$ is linear. *Step 2.* Let $u(c) := 1 - e^{-c^2/2}$ and $g := \sqrt{u}$. Then g is nondecreasing and $g'' = (2uu'' - u'^2)/(4u^{3/2})$ on $(0, \infty)$, with $u' = ce^{-c^2/2}$ and $u'' = (1-c^2)e^{-c^2/2}$. With $a := c^2/2$, a direct computation gives $2uu'' - u'^2 = 2e^{-a}\,\kappa(a)$ where $\kappa(a) := 1 - e^{-a} - 2a + ae^{-a}$. Since $\kappa(0) = 0$ and $\kappa'(a) = -2(1-e^{-a}) - ae^{-a} < 0$ for $a > 0$, $\kappa < 0$ on $(0, \infty)$, hence $g'' < 0$ and g is concave on $[0, \infty)$ (it is continuous at 0). *Step 3.* The minimum of two concave nondecreasing functions is concave and nondecreasing, and $g \le 1$. □

Theorem 1 (Total-variation transfer)**.** *Let $\psi : [0, +\infty] \to [0, 1]$ be concave and nondecreasing, and let the cost satisfy $\mathrm{TV}(P^v, P^{(i)}) \le \psi(c(v, v_i))$ for every $v \in \mathcal{V}$ and $i \in [\![M]\!]$. If $S_c^w(v; \mathcal{D}) \le \varepsilon$, then*

$$\mathrm{TV}(P^v, \bar{P}_w) \le \psi\big((M+1)\varepsilon\big) \quad \text{and} \quad P^v(A) \le \bar{P}_w(A) + \psi\big((M+1)\varepsilon\big) \text{ for every event } A.$$

By Lemma 4, the costs $c = \sqrt{2\mathrm{KL}(P^v \| P^{(i)})}$ (reverse) and $c = \sqrt{2\mathrm{KL}(P^{(i)} \| P^v)}$ (forward) satisfy the hypothesis with $\psi = \varphi$.

Proof: Step 1 (convexity of TV *in its second argument).* For every event A, $|P^v(A) - \bar{P}_w(A)| = |\sum_i w_i (P^v(A) - P^{(i)}(A))| \le \sum_i w_i |P^v(A) - P^{(i)}(A)| \le \sum_i w_i \mathrm{TV}(P^v, P^{(i)})$; taking the supremum over A, $\mathrm{TV}(P^v, \bar{P}_w) \le \sum_i w_i \mathrm{TV}(P^v, P^{(i)})$. *Step 2 (hypothesis).* $\sum_i w_i \mathrm{TV}(P^v, P^{(i)}) \le \sum_i w_i \psi(c(v, v_i))$. *Step 3 (Jensen).* Since ψ is concave and w is a probability vector, $\sum_i w_i \psi(c(v, v_i)) \le \psi(\sum_i w_i c(v, v_i))$. *Step 4 (region).* By (3), $\sum_i w_i c(v, v_i) \le (M+1)\varepsilon$, and ψ is nondecreasing, so $\psi(\sum_i w_i c(v, v_i)) \le \psi((M+1)\varepsilon)$. *Step 5 (events).* By the definition of TV, $P^v(A) - \bar{P}_w(A) \le \mathrm{TV}(P^v, \bar{P}_w)$. □

Theorem 2 (Rare events stay rare)**.** *Let $c(v, v_i) := \mathrm{KL}(P^v \| P^{(i)})$ (reverse divergence, cost not square-rooted). If $S_c^w(v; \mathcal{D}) \le \varepsilon$, then (i) $\mathrm{KL}(P^v \| \bar{P}_w) \le (M+1)\varepsilon$, and (ii) for every event A with $q := \bar{P}_w(A) \in (0, 1)$,*

$$P^v(A) \;\le\; \mathrm{kl}^{-1}\big(q, (M+1)\varepsilon\big) := \sup\big\{p \in [q, 1] : \mathrm{kl}(p\|q) \le (M+1)\varepsilon\big\} \;\le\; \frac{(M+1)\varepsilon + \log 2}{\log(1/q)}.$$

Proof: Step 1 (convexity of KL *in its second argument).* By joint convexity of the relative entropy (Cover and Thomas 2006, Thm. 2.7.2) applied to the pairs $(P^v, P^{(i)})$ with weights w_i, $\mathrm{KL}(P^v \| \sum_i w_i P^{(i)}) \le \sum_i w_i \mathrm{KL}(P^v \| P^{(i)})$. *Step 2 (region).* By (3), $\sum_i w_i \mathrm{KL}(P^v \| P^{(i)}) \le (M+1)\varepsilon$. This proves (i). *Step 3 (data processing).* The map $\mathbf{x}_{0:J} \mapsto \mathbb{1}_A(\mathbf{x}_{0:J})$ sends P^v to a Bernoulli law with parameter $P^v(A)$ and $\bar{P}_w$ to a Bernoulli law with parameter q. The data-processing inequality for KL gives $\mathrm{kl}(P^v(A)\|q) \le \mathrm{KL}(P^v \| \bar{P}_w) \le (M+1)\varepsilon$. *Step 4 (inversion).* The map $p \mapsto \mathrm{kl}(p\|q)$ is continuous on $[0, 1]$, vanishes at $p = q$ and has derivative $\log \frac{p(1-q)}{q(1-p)} > 0$ on $(q, 1)$. If $P^v(A) \le q$, the first bound holds because $\mathrm{kl}^{-1}(q, \cdot) \ge q$. Otherwise $P^v(A) \in (q, 1]$ belongs to the set whose supremum defines $\mathrm{kl}^{-1}(q, (M+1)\varepsilon)$, by Step 3. *Step 5 (closed form).* For $p \in [0, 1]$, $\mathrm{kl}(p\|q) = p \log \frac{1}{q} + (1-p) \log \frac{1}{1-q} - \mathrm{h}(p)$ with the binary entropy $\mathrm{h}(p) = -p \log p - (1-p) \log(1-p) \le \log 2$. The middle term is nonnegative, so $\mathrm{kl}(p\|q) \ge p \log \frac{1}{q} - \log 2$. Every p with $\mathrm{kl}(p\|q) \le (M+1)\varepsilon$ therefore satisfies $p \le ((M+1)\varepsilon + \log 2)/\log(1/q)$. □

Proposition 2 (Forward and reverse costs of an unsafe mode)**.** *Let P, Q be probability measures on a common space, $m \in [0, 1)$ and $P' := (1-m)P + mQ$ (a safe law P contaminated by a component Q). Then: (i) $\mathrm{KL}(P\|P') \le \log \frac{1}{1-m}$; (ii) for every event A with $q := P(A) \in (0, 1)$ and $p := P'(A)$, $\mathrm{KL}(P'\|P) \ge \mathrm{kl}(p\|q) \ge p \log \frac{p}{q} - p$. If $Q(A) = 1$ and $m \ge q$, then $\mathrm{KL}(P'\|P) \ge m \log \frac{m}{q} - m$.*

Proof: (i) *Step 1.* For every measurable B, $P'(B) \ge (1-m)P(B)$, so $P \ll P'$ and $\frac{dP}{dP'} \le \frac{1}{1-m}$ P'-almost everywhere. *Step 2.* $\mathrm{KL}(P\|P') = \mathbb{E}_P[\log \frac{dP}{dP'}] \le \log \frac{1}{1-m}$. (ii) *Step 1.* As in Step 3 of Theorem 2, $\mathrm{KL}(P'\|P) \ge \mathrm{kl}(p\|q)$. *Step 2.* Since $\log \frac{1}{1-q} \ge 0$, $\mathrm{kl}(p\|q) \ge p \log \frac{p}{q} + (1-p)\log(1-p)$, and $(1-p)\log(1-p) \ge -p$ because $\log(1-p) \ge -\frac{p}{1-p}$ for $p < 1$ (the term is 0 at $p = 1$). *Step 3.* If $Q(A) = 1$, then $p = (1-m)q + m \ge \max\{q, m\}$. The map $x \mapsto x \log \frac{x}{q} - x$ has derivative $\log \frac{x}{q} \ge 0$ on $[q, 1]$, so for $m \ge q$ its value at p is at least its value at m. □

Proposition 2 is the reason why the direction of the divergence matters. A forward cost, measured along the historical samplers, charges at most about m for an unsafe mode of mass m, whatever the rarity q of the unsafe region under the safe samplers. This is the same order as the variation between legitimate tasks, so a region calibrated to contain legitimate tasks also contains such contaminated fields. The reverse cost, measured along the candidate's own trajectories, grows like $m \log(m/q)$ and becomes large exactly when the unsafe region is rare. For the chains of the experiments, the contaminated field does not generate exactly the mixture $(1-m)P + mQ$ (the chains start from $\mathcal{N}(0, I_n)$ and are discretised). The measured forward costs nevertheless stay within a factor of three of $\log \frac{1}{1-m}$, while the reverse costs exceed it by a factor of 9 to 20 with exact scores and 3 to 13 with linear fits (Table 1).

Remark 3 (Monte Carlo)**.** *Theorems 1 and 2 are stated for the exact costs. In the experiments the costs are Monte Carlo estimates on 256 frozen probe trajectories per field. The coverage of Proposition 1 is exact for the estimated score, but the certificates of Section 7 are* plug-in*: they assume that the estimated region is contained in the exact one. A fully rigorous certificate replaces the estimate of $\mathrm{KL}(P^v\|P^{(i)})$ by an upper confidence bound computed on fresh trajectories. The historical event probability $\bar{P}(A)$ is replaced by a 95% Clopper–Pearson upper bound.*

Remark 4 (The order-1 floor)**.** *For a metric cost d and every v, the triangle inequality $d(v_i, v_j) \le d(v_i, v) + d(v, v_j)$ summed over $i, j \in [\![M]\!]$ gives*

$$\frac{1}{M}\sum_{i=1}^{M} d(v, v_i) \;\ge\; \frac{1}{2M^2}\sum_{i,j=1}^{M} d(v_i, v_j).$$

The region is therefore non-empty only if the budget $(M+1)\varepsilon$ is at least half the mean pairwise distance between the historical fields, and the bounds of Theorems 1 and 2 are informative only when the historical safe fields are close to each other in the chosen divergence. This is the vector-field counterpart of the diameter-order behaviour of the order-1 estimator discussed in the Hausdorff note (`pr:order_one`). The weighted score S_c^w, with weights concentrated on the historical tasks most similar to the new one, is the natural way to shrink the effective spread.

Proposition 3 (Deterministic samplers: Wasserstein transfer)**.** *Consider ODE chains $X_{k+1}^v = X_k^v + h_k v(X_k^v, t_k)$, all started from the same $X_0 \sim \mu_0$. Assume that $v(\cdot, t_k)$ is L_k-Lipschitz for each k, and let $\Gamma_k := \prod_{l=k+1}^{J-1}(1 + h_l L_l)$ (empty product $= 1$). Define*

$$c_W(v, v_i) := \sum_{k=0}^{J-1} h_k \Gamma_k \;\mathbb{E}\left\|v(X_k^{v_i}, t_k) - v_i(X_k^{v_i}, t_k)\right\|.$$

If $S_{c_W}^w(v; \mathcal{D}) \le \varepsilon$, then for every $r > 0$ and every Borel set $B \subseteq \mathbb{R}^n$, with $B^{+r} := \{\mathbf{y} : \operatorname{dist}(\mathbf{y}, B) < r\}$,

$$\mathbb{P}(X_J^v \in B) \;\le\; \sum_i w_i \,\mathbb{P}(X_J^{v_i} \in B^{+r}) + \frac{(M+1)\varepsilon}{r}.$$

Proof: Step 1 (one step). Let $e_k := \|X_k^v - X_k^{v_i}\|$ and $\delta_k := \|v(X_k^{v_i}, t_k) - v_i(X_k^{v_i}, t_k)\|$. Adding and subtracting $h_k v(X_k^{v_i}, t_k)$, $X_{k+1}^v - X_{k+1}^{v_i} = (X_k^v - X_k^{v_i}) + h_k[v(X_k^v, t_k) - v(X_k^{v_i}, t_k)] + h_k[v(X_k^{v_i}, t_k) - v_i(X_k^{v_i}, t_k)]$, so by the Lipschitz property $e_{k+1} \le (1 + h_k L_k)e_k + h_k\delta_k$. *Step 2 (discrete Grönwall).* Since $e_0 = 0$, induction on k gives $e_J \le \sum_{k=0}^{J-1} h_k \Gamma_k \delta_k$, hence $\mathbb{E}\, e_J \le c_W(v, v_i)$. *Step 3 (a coupling with the mixture).* Draw $I \sim w$ independently of X_0 and run the chains of v and v_I from the same X_0. The pair $(X_J^v, X_J^{v_I})$ has marginals $\operatorname{law}(X_J^v)$ and $\sum_i w_i \operatorname{law}(X_J^{v_i})$, and by Step 2 and (3), $\mathbb{E}\,\|X_J^v - X_J^{v_I}\| = \sum_i w_i \,\mathbb{E}\,\|X_J^v - X_J^{v_i}\| \le \sum_i w_i c_W(v, v_i) \le (M+1)\varepsilon$. *Step 4 (events).* If $X_J^v \in B$ and $\|X_J^v - X_J^{v_I}\| < r$, then $\operatorname{dist}(X_J^{v_I}, B) < r$, i.e. $X_J^{v_I} \in B^{+r}$. Hence $\mathbb{P}(X_J^v \in B) \le \mathbb{P}(X_J^{v_I} \in B^{+r}) + \mathbb{P}(\|X_J^v - X_J^{v_I}\| \ge r)$, and Markov's inequality bounds the last term by $(M+1)\varepsilon/r$. □

For a linear field $v_{\boldsymbol{\theta}} = \sum_j \theta_j \phi_j$ with columns ϕ_j of Φ satisfying $\|\nabla_{\mathbf{x}} \phi_j(\mathbf{x}, t)\|_{\mathrm{op}} \le \ell_j(t)$, the mean-value theorem, the triangle inequality and the Cauchy–Schwarz inequality give $L(t) \le \sum_j |\theta_j| \ell_j(t) \le \|\boldsymbol{\theta}\|_2 (\sum_j \ell_j(t)^2)^{1/2}$, so Γ_k is explicit. It is also astronomically large: for the centre field of the experiments, a bound of the first type (grouping the two output coordinates of each feature) gives $L(t_k) \in [52, 53]$ and $\Gamma_0 \approx 10^{15}$. Proposition 3 is valid but vacuous in practice; this is the reason for sampling with the SDE chain when a certificate is wanted, see also Benton et al. (2024) for Grönwall-type bounds in flow matching.

6 Linear flow matching

Definition 4 (Linear flow matching)**.** *Let $\Phi : \mathbb{R}^n \times [0,1] \to \mathbb{R}^{n \times D}$ be a measurable feature map and $v_{\boldsymbol{\theta}}(\mathbf{x}, t) := \Phi(\mathbf{x}, t)\boldsymbol{\theta}$, $\boldsymbol{\theta} \in \mathbb{R}^D$. For a data law p, let $t \sim \mathcal{U}[0,1]$, $\mathbf{x}_0 \sim \mathcal{N}(0, I_n)$, $\mathbf{x}_1 \sim p$ be independent and $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$. The flow-matching moments and loss are*

$$A_p := \mathbb{E}\left[\Phi(\mathbf{x}_t, t)^\top \Phi(\mathbf{x}_t, t)\right], \quad \mathbf{b}_p := \mathbb{E}\left[\Phi(\mathbf{x}_t, t)^\top (\mathbf{x}_1 - \mathbf{x}_0)\right], \quad L_p(\boldsymbol{\theta}) := \mathbb{E}\,\|\Phi(\mathbf{x}_t, t)\boldsymbol{\theta} - (\mathbf{x}_1 - \mathbf{x}_0)\|^2,$$

and the ridge fit is $\boldsymbol{\theta}_p := \arg\min_{\boldsymbol{\theta}} L_p(\boldsymbol{\theta}) + \lambda\|\boldsymbol{\theta}\|^2 = (A_p + \lambda I)^{-1}\mathbf{b}_p$, $\lambda > 0$.

Lemma 5 (Moments are linear in the data law)**.** *$L_p(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top A_p \boldsymbol{\theta} - 2\mathbf{b}_p^\top \boldsymbol{\theta} + \mathbb{E}\,\|\mathbf{x}_1 - \mathbf{x}_0\|^2$, and for $p' = (1-m)p + mq$: $A_{p'} = (1-m)A_p + mA_q$ and $\mathbf{b}_{p'} = (1-m)\mathbf{b}_p + m\mathbf{b}_q$.*

Proof: Expand the square. Conditioning on $\mathbf{x}_1$, each moment is an integral of a function of $\mathbf{x}_1$ against p, which is linear in p. □

In the experiments, Lemma 5 gives the fit of a contaminated task exactly from the moments of the clean task and of the contaminating component, and the few-shot moments $(\hat{A}, \hat{\mathbf{b}})$ are those of the empirical law of the N new samples (with many draws of $(t, \mathbf{x}_0)$ per sample).

Proposition 4 (Field distance is a Mahalanobis distance)**.** *Let ρ be a probability measure on $\mathbb{R}^n \times [0,1]$ with $\mathbb{E}_\rho\,\|\Phi\|_F^2 < \infty$ and $G_\rho := \mathbb{E}_\rho[\Phi^\top \Phi] \succeq 0$. (a) For all $\boldsymbol{\theta}, \boldsymbol{\theta}'$, $\|v_{\boldsymbol{\theta}} - v_{\boldsymbol{\theta}'}\|_{L^2(\rho)}^2 := \mathbb{E}_{(\mathbf{x},t)\sim\rho}\,\|v_{\boldsymbol{\theta}}(\mathbf{x}, t) - v_{\boldsymbol{\theta}'}(\mathbf{x}, t)\|^2 = (\boldsymbol{\theta} - \boldsymbol{\theta}')^\top G_\rho (\boldsymbol{\theta} - \boldsymbol{\theta}')$. Hence $d_\rho(\boldsymbol{\theta}, \boldsymbol{\theta}') := \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_{G_\rho}$ is a pseudometric on $\mathbb{R}^D$; it is a metric if and only if $G_\rho \succ 0$. (b) If $\rho = \frac{1}{M}\sum_i \rho_i$, where ρ_i is the law of $(\mathbf{x}_t, t)$ under task i (the pooled interpolation path of the training tasks), then $G_\rho = \frac{1}{M}\sum_i A_{p_i}$. (c) For $\delta > 0$ let $G_\delta := G_\rho + \delta I = CC^\top$ (Cholesky). With $\mathbf{z} := C^\top \boldsymbol{\theta}$, $\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_{G_\delta} = \|\mathbf{z} - \mathbf{z}'\|_2$, so the order-1 score with cost $d_\delta = \|\cdot\|_{G_\delta}$ is the Euclidean score of the manuscript in whitened coordinates: $S_{d_\delta}(\boldsymbol{\theta}; \mathcal{D}) = \frac{1}{M(M+1)}\sum_i \|\mathbf{z} - \mathbf{z}_i\|_2$. (d) If $\Phi' = \Phi T$ with T invertible and $\boldsymbol{\theta}' = T^{-1}\boldsymbol{\theta}$, then $v'_{\boldsymbol{\theta}'} = v_{\boldsymbol{\theta}}$ and $G'_\rho = T^\top G_\rho T$, so $d'_\rho(T^{-1}\boldsymbol{\theta}_1, T^{-1}\boldsymbol{\theta}_2) = d_\rho(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$.*

Proof: (a) Pointwise, $\|\Phi(\boldsymbol{\theta}-\boldsymbol{\theta}')\|^2 = (\boldsymbol{\theta}-\boldsymbol{\theta}')^\top \Phi^\top \Phi(\boldsymbol{\theta}-\boldsymbol{\theta}')$; take the expectation under ρ. The map $\mathbf{a} \mapsto (\mathbf{a}^\top G_\rho \mathbf{a})^{1/2}$ is a seminorm, so d_ρ is a pseudometric, and it separates points if and only if G_ρ has a trivial kernel. (b) The expectation under a mixture is the mixture of expectations, and $\mathbb{E}_{\rho_i}[\Phi^\top \Phi] = A_{p_i}$ by Definition 4. (c) $(\boldsymbol{\theta}-\boldsymbol{\theta}')^\top CC^\top (\boldsymbol{\theta}-\boldsymbol{\theta}') = \|C^\top(\boldsymbol{\theta}-\boldsymbol{\theta}')\|_2^2$. (d) $\Phi'\boldsymbol{\theta}' = \Phi T T^{-1}\boldsymbol{\theta} = \Phi\boldsymbol{\theta}$, and $\mathbb{E}_\rho[\Phi'^\top \Phi'] = T^\top G_\rho T$; substitute in (a). □

Three remarks. First, G_ρ depends only on the training tasks, so the region keeps the coverage of Proposition 1. Second, the regularisation δ adds $\delta\|\boldsymbol{\theta}-\boldsymbol{\theta}'\|^2$ to the squared distance and turns the pseudometric into a metric; the experiments use $\delta = 10^{-6}\,\mathrm{tr}(G_\rho)/D$. Third, for linear models there are no hidden units to permute, so the weight-alignment step that the manuscript needs for neural networks is unnecessary, and (d) shows that the distance does not depend on how the features are parametrised.

Proposition 5 (`OT1-TRAIN` is a second-order cone program)**.** *Given few-shot moments $(\hat{A}, \hat{\mathbf{b}})$ with $\hat{A} \succeq 0$, $\lambda > 0$, anchors $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_M$, $G_\delta = CC^\top$ and a budget $R \geq 0$, consider*

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^D} \ \boldsymbol{\theta}^\top(\hat{A}+\lambda I)\boldsymbol{\theta} - 2\hat{\mathbf{b}}^\top\boldsymbol{\theta} \quad \textit{s.t.} \quad \sum_{i=1}^{M} \|C^\top(\boldsymbol{\theta}-\boldsymbol{\theta}_i)\|_2 \leq R. \tag{4}$$

With $R = M(M+1)\varepsilon$ the feasible set is the region $\{\boldsymbol{\theta} : S_{d_\delta}(\boldsymbol{\theta}; \mathcal{D}) \leq \varepsilon\}$. (i) Problem (4) *is a convex second-order cone program. (ii) If the feasible set is non-empty, it has a unique minimiser. The feasible set is non-empty if and only if $S_{d_\delta}(\boldsymbol{\theta}_c; \mathcal{D}) \leq \varepsilon$, where $\boldsymbol{\theta}_c$ is a geometric median of the anchors in the metric d_δ.*

Proof: (i) The objective is a convex quadratic. With $\mathbf{z} = C^\top\boldsymbol{\theta}$ and epigraph variables $u_i \geq \|\mathbf{z} - \mathbf{z}_i\|_2$, the constraint becomes M second-order cones and the linear inequality $\sum_i u_i \leq R$; the quadratic objective is represented by one rotated second-order cone (Boyd and Vandenberghe 2004). (ii) The feasible set is closed and convex, and the objective is strictly convex and coercive because $\hat{A} + \lambda I \succ 0$, so a minimiser exists and is unique. The left-hand side of the constraint is minimised exactly at the geometric medians, which gives the last claim. □

The geometric median is computed by Weiszfeld's iteration (Weiszfeld 1937) in the whitened coordinates; it lies in the convex hull of the anchors. It is the field of lowest score, the vector-field counterpart of the model-fusion corollary of the manuscript, and we call it the *centre* of the region.

On-policy costs. The fixed Gram G_ρ measures field differences on the pooled historical path. The on-policy variants measure them along the candidate's own trajectories:

- ODE sampler: $G_{\mathrm{on}}(\boldsymbol{\theta}) \coloneqq \frac{1}{J}\sum_k \mathbb{E}_{P^{\boldsymbol{\theta}}}[\Phi(X_k,t_k)^\top \Phi(X_k,t_k)]$ (plus δI) and the cost $\|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|_{G_{\mathrm{on}}(\boldsymbol{\theta})}$;
- SDE sampler: the cost $\mathrm{KL}(P^{\boldsymbol{\theta}}\|P^{(i)}) = \frac{1}{2}\|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|^2_{G_{\mathrm{KL}}(\boldsymbol{\theta})}$ of Corollary 1(iii), to which Theorem 2 applies.

The forward SDE cost $\mathrm{KL}(P^{(i)}\|P^{\boldsymbol{\theta}}) = \frac{1}{2}\|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|^2_{G_{\mathrm{KL}}(\boldsymbol{\theta}_i)}$ uses the Gram matrix of the historical chain i and is therefore convex in $\boldsymbol{\theta}$. The on-policy costs are not, because the Gram matrix depends on $\boldsymbol{\theta}$. Two elementary facts make them tractable.

Lemma 6 (A frozen-Gram KL region is an ellipsoid around the parameter average)**.** *Let $G \succeq 0$, $\bar{\boldsymbol{\theta}} \coloneqq \frac{1}{M}\sum_i \boldsymbol{\theta}_i$ and $\varepsilon \geq 0$. Then $\sum_i \|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|_G^2 = M\|\boldsymbol{\theta}-\bar{\boldsymbol{\theta}}\|_G^2 + \sum_i \|\boldsymbol{\theta}_i-\bar{\boldsymbol{\theta}}\|_G^2$, and therefore*

$$\Big\{\boldsymbol{\theta} : \frac{1}{M(M+1)}\sum_i \tfrac{1}{2}\|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|_G^2 \leq \varepsilon\Big\} = \Big\{\boldsymbol{\theta} : \|\boldsymbol{\theta}-\bar{\boldsymbol{\theta}}\|_G^2 \leq 2(M+1)\varepsilon - \frac{1}{M}\sum_i \|\boldsymbol{\theta}_i-\bar{\boldsymbol{\theta}}\|_G^2\Big\}.$$

Proof: Write $\boldsymbol{\theta}-\boldsymbol{\theta}_i = (\boldsymbol{\theta}-\bar{\boldsymbol{\theta}}) + (\bar{\boldsymbol{\theta}}-\boldsymbol{\theta}_i)$ and expand the square; the cross terms sum to $2\langle \boldsymbol{\theta}-\bar{\boldsymbol{\theta}}, \sum_i(\bar{\boldsymbol{\theta}}-\boldsymbol{\theta}_i)\rangle_G = 0$. Multiply the constraint by $2M(M+1)$ and divide by M. □

So, with the Gram frozen, the KL region is an ellipsoid centred at the parameter average (a "model soup"), whereas the metric region of Proposition 5 is centred at the geometric median.

Proposition 6 (Radial back-off)**.** *Let $S : \mathbb{R}^D \to [0,+\infty]$ be any score, $\varepsilon \geq 0$, $\boldsymbol{\theta}_c$ with $S(\boldsymbol{\theta}_c) \leq \varepsilon$ and $\boldsymbol{\theta} \in \mathbb{R}^D$. Run the bisection:* $\mathrm{lo} \leftarrow 0$, $\mathrm{hi} \leftarrow 1$*; repeat* $\mathrm{mid} \leftarrow (\mathrm{lo}+\mathrm{hi})/2$*; if $S((1-\mathrm{mid})\boldsymbol{\theta}+\mathrm{mid}\,\boldsymbol{\theta}_c) \leq \varepsilon$ then* $\mathrm{hi} \leftarrow \mathrm{mid}$*, else* $\mathrm{lo} \leftarrow \mathrm{mid}$*. After any number of iterations, $\boldsymbol{\theta}_{\mathrm{hi}} \coloneqq (1-\mathrm{hi})\boldsymbol{\theta} + \mathrm{hi}\,\boldsymbol{\theta}_c$ satisfies $S(\boldsymbol{\theta}_{\mathrm{hi}}) \leq \varepsilon$. If moreover S is convex and $S(\boldsymbol{\theta}_c) < \varepsilon < S(\boldsymbol{\theta})$, then $\lambda^\star \coloneqq \frac{S(\boldsymbol{\theta})-\varepsilon}{S(\boldsymbol{\theta})-S(\boldsymbol{\theta}_c)} \in (0,1)$ gives $S((1-\lambda^\star)\boldsymbol{\theta} + \lambda^\star\boldsymbol{\theta}_c) \leq \varepsilon$.*

Proof: Invariant. "$\boldsymbol{\theta}_{\mathrm{hi}}$ is feasible" holds initially, since $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_c$, and is preserved, since hi is only ever replaced by a feasible mid. No continuity of S is needed. *Convex case.* $S((1-\lambda)\boldsymbol{\theta}+\lambda\boldsymbol{\theta}_c) \leq (1-\lambda)S(\boldsymbol{\theta}) + \lambda S(\boldsymbol{\theta}_c) = S(\boldsymbol{\theta}) - \lambda(S(\boldsymbol{\theta}) - S(\boldsymbol{\theta}_c))$, which equals ε at $\lambda = \lambda^\star$. □

For a linear model, blending parameters blends fields: $v_{(1-\lambda)\boldsymbol{\theta}+\lambda\boldsymbol{\theta}_c} = (1-\lambda)v_{\boldsymbol{\theta}} + \lambda v_{\boldsymbol{\theta}_c}$. The back-off is therefore also an inference-time safety filter (evaluate both fields and mix them), and it is the radial retraction of the gauge parametrisation of the companion note on free parametrisation, applied in function space.

Algorithm 1 Few-shot training with an on-policy order-1 constraint (sequential convex programming).

Require: few-shot moments $(\hat{A}, \hat{\mathbf{b}})$; anchors $\boldsymbol{\theta}_{1:M}$; centre $\boldsymbol{\theta}_c$ with $S_{\mathrm{on}}(\boldsymbol{\theta}_c) < \varepsilon_{\mathrm{on}}$; iterations I; frozen probe noise
1: $\boldsymbol{\theta}^{(0)} \leftarrow$ solution of (4) with the fixed Gram G_δ and $R = M(M+1)\varepsilon_{\mathrm{ref}}$
2: **for** $j = 0, \dots, I-1$ **do**
3: simulate the chain of $\boldsymbol{\theta}^{(j)}$ on the probe noise; form $G_{\mathrm{on}}(\boldsymbol{\theta}^{(j)}) = C_j C_j^\top$
4: $\boldsymbol{\theta}^{(j+1)} \leftarrow$ solution of (4) with $C = C_j$ and $R = M(M+1)\varepsilon_{\mathrm{on}}$
5: **end for**
6: **if** $S_{\mathrm{on}}(\boldsymbol{\theta}^{(I)}) > \varepsilon_{\mathrm{on}}$ **then** radial back-off of $\boldsymbol{\theta}^{(I)}$ towards $\boldsymbol{\theta}_c$ (Proposition 6)
7: **end if**
8: **return** the resulting $\boldsymbol{\theta}$, which satisfies $S_{\mathrm{on}}(\boldsymbol{\theta}) \le \varepsilon_{\mathrm{on}}$

Table 1: Divergence between a contaminated field and its own clean field (median over the pool), and probability of the unsafe set. Proposition 2 bounds the forward divergence by $\log \frac{1}{1-m}$ for exact mixtures, and the measured values stay within a factor of three of it; the reverse divergence exceeds it by a factor of 9–20 (A) and 3–13 (D). Safe fields: $P^v(A)$ averages 8.3×10^{-6} (A) and 0.0004 (D), and the median pairwise divergence between safe fields is 0.067 (A) and 0.378 (D) nats.

		(A) diffusion, exact scores			(D) linear FM, SDE		
m	$\log \frac{1}{1-m}$	$P^v(A)$	forward KL	reverse KL	$P^v(A)$	forward KL	reverse KL
0.002	0.0020	0.0030	0.0043	0.039	–	–	–
0.01	0.0101	0.0121	0.0253	0.142	0.0018	0.021	0.029
0.03	0.0305	0.0313	0.0677	0.441	0.0096	0.087	0.207
0.1	0.1054	0.0941	0.1840	1.039	0.0635	0.273	1.309
0.3	0.3567	0.2654	0.4810	3.273	0.2456	0.730	4.681

7 Toy experiment

7.1 Setup

Tasks and unsafe set. In $\mathbb{R}^2$, a task $\tau = (\phi, \varrho)$ is the equal-weight mixture of four Gaussians $\mathcal{N}(\varrho(\cos(\phi + j\pi/2), \sin(\phi + j\pi/2)), 0.3^2 I_2)$, $j = 0, \dots, 3$, with $\phi \sim \mathcal{N}(0, 0.05^2)$ and $\varrho \sim \mathcal{N}(2, 0.05^2)$. The family is concentrated: the median path divergence between two tasks is 0.067 nats (Table 1). The unsafe event is $A = \{\|\mathbf{x}_J\| \le 0.6\}$, a disk at the centre that no task visits (Figure 1). A contaminated field of mass m is the field of $(1-m)p_\tau + m\,\mathcal{N}(0, 0.25^2 I_2)$. The method never uses A; it is only used for evaluation.

Fields and samplers. *Pilot A (diffusion, exact scores)*: the fields are the exact scores of the noised mixtures (closed form for Gaussian mixtures), sampled with the diffusion chain of Section 4 on the grid $t_k = t_{\min} + (T - t_{\min})(1 - k/J)^2$, $T = 6$, $t_{\min} = 0.01$, $J = 50$. There is no estimation error, which isolates the choice of the cost. *Pilots C and D (linear flow matching)*: $\Phi(\mathbf{x}, t) = I_2 \otimes \psi(\mathbf{x}, t)^\top$ with $\psi(\mathbf{x}, t) = \big(\sqrt{2/F}\cos(\boldsymbol{\omega}_j^\top \mathbf{x} + \nu_j t + b_j)\big)_{j \le F} \oplus (1, t, \mathbf{x}, t\mathbf{x})$, $F = 256$ random Fourier features (Rahimi and Recht 2007) ($\boldsymbol{\omega}_j = s_j \mathbf{g}_j$, $\mathbf{g}_j \sim \mathcal{N}(0, I_2)$, s_j uniform on $\{0.6, 1.2, 2.4, 4.8\}$, $\nu_j \sim \mathcal{N}(0, 9)$, $b_j \sim \mathcal{U}[0, 2\pi]$), so $D = 2 \times 262 = 524$. Each pipeline output is the ridge fit of Definition 4 ($\lambda = 10^{-5}$, 20 000 Monte Carlo samples per task). Pilot C uses the ODE sampler, pilot D the SDE sampler with $\eta = 1$, both on the uniform grid with $J = 50$.

Regions and splits. A pool of 200 tasks is split at random into $M = 20$ training, $K = 40$ calibration and 140 test tasks; $\beta = 0.1$, so the nominal coverage is $k/(K+1) = 37/41 = 0.902$. Pilots A and D use 200 random splits and pilot C uses 100. All scores use 256 probe trajectories per field with frozen noise. Event probabilities are estimated with 20 000 (A), 10 000 (C) and 4 000 (D) samples.

Few-shot fine-tuning (pilot C). On one fixed split, for each of the first 32 test tasks, $N = 12$ clean samples are drawn, together with 0 or 3 samples from $\mathcal{N}(0, 0.25^2 I_2)$ (20% contamination). The estimators are: least squares $(\hat{A} + \lambda I)^{-1}\hat{\mathbf{b}}$ for $\lambda \in \{10^{-5}, 10^{-3}, 10^{-1}\}$; shrinkage towards the centre, $\arg\min_{\boldsymbol{\theta}} \boldsymbol{\theta}^\top(\hat{A} + \lambda I)\boldsymbol{\theta} - 2\hat{\mathbf{b}}^\top\boldsymbol{\theta} + \gamma\|\boldsymbol{\theta} - \boldsymbol{\theta}_c\|^2_{G_\rho}$ for $\gamma \in \{0.1, 1, 10\}$; the conformal SOCP (4) with the fixed Gram ($\varepsilon_{\mathrm{ref}} = 0.0182$), solved with CVXPY (Diamond and Boyd 2016) and Clarabel (Goulart and Chen 2024); Algorithm 1 with $I = 4$ and 16 bisection steps ($\varepsilon_{\mathrm{on}} = 0.0193$); the centre $\boldsymbol{\theta}_c$ alone; and the ample-data fit $f(\tau)$ as an oracle. Quality is the unbiased squared maximum mean discrepancy (Gretton et al. 2012) between 2 500 generated and 2 500 clean task samples, with kernel $\sum_{s \in \{0.1, 0.3, 1\}} e^{-\|\mathbf{x} - \mathbf{y}\|^2/(2s^2)}$.

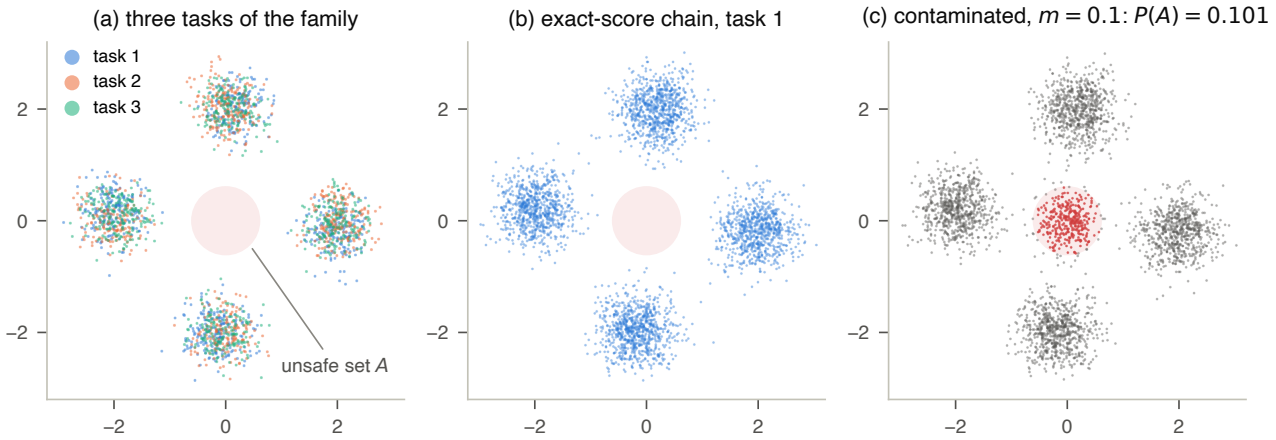


Figure 1: Task family and unsafe set. (a) Samples of three tasks. (b) Output of the exact-score diffusion chain of task 1, which reproduces the task. (c) Output of the chain of the contaminated field with $m = 0.1$; samples in the unsafe set A are red.

Table 2: Mean coverage of held-out safe fields (nominal 0.902) and fraction of contaminated fields outside the region, by unsafe mass m, averaged over random splits.

cost $c(v, v_i)$	coverage	$m = 0.002$	0.01	0.03	0.1	0.3
(a) diffusion, exact scores (200 splits)						
forward, $c = \sqrt{2\mathrm{KL}(P^{(i)} \Vert P^v)}$	0.897	0.13	0.19	0.42	0.98	1.00
reverse, $c = \sqrt{2\mathrm{KL}(P^v \Vert P^{(i)})}$	0.897	0.25	0.93	1.00	1.00	1.00
reverse, $c = \mathrm{KL}(P^v \Vert P^{(i)})$	0.898	0.20	0.83	1.00	1.00	1.00
(b) linear flow matching, ODE sampler (100 splits)						
fixed Gram, $c = \Vert\theta - \theta_i\Vert_G$	0.902	–	0.10	0.15	0.87	1.00
on-policy Gram, $c = \Vert\theta - \theta_i\Vert_{G_{\mathrm{on}}(\theta)}$	0.899	–	0.13	0.55	1.00	1.00
(c) linear flow matching, SDE sampler (200 splits)						
forward, $c = \mathrm{KL}(P^{(i)} \Vert P^v)$	0.900	–	0.10	0.17	0.69	1.00
reverse, $c = \mathrm{KL}(P^v \Vert P^{(i)})$	0.899	–	0.13	0.55	1.00	1.00

7.2 The direction of the cost decides what the region excludes

Table 2 and Figure 2 report coverage and exclusion. Coverage is at the nominal level for every cost, as Proposition 1 guarantees. Exclusion depends strongly on the direction of the cost. With exact scores, the forward region excludes only 0.42 of the fields that carry 3% unsafe mass, against 1.00 for the reverse-KL region, and at $m = 0.01$ the rates are 0.19 against 0.83. The linear flow-matching pilots show the same ordering (0.15 against 0.55 for the ODE sampler, 0.17 against 0.55 for the SDE sampler at $m = 0.03$), with a smaller gap. This is consistent with the larger variation between the fitted safe fields (median pairwise divergence 0.378 against 0.067 nats for exact scores), which widens the calibrated budget. Figure 3 shows the mechanism on one split: under the forward cost, the contaminated fields have scores that overlap those of the safe fields, while under the reverse cost they are separated. Figure 4 shows where in time the reverse divergence accumulates: at the late, near-data steps, where the contaminated trajectories are pulled into A and the safe fields push them out.

7.3 Certificates

Table 3 and Figure 5 check the certificates of Theorems 1 and 2 with the plug-in convention of Remark 3. No split violates a bound. The total-variation certificate is valid but loose (0.283 with exact scores, 0.553 with linear fits), because total variation cannot use the rarity of A. The reverse-KL certificate of Theorem 2 is 0.031 with exact scores, within a factor of three of the worst field actually inside the region (0.0123), and 0.134 with linear fits (worst field 0.0145). The difference between the two pilots is the floor of Remark 4: the linear fits are more spread out in divergence, so the calibrated budget $(M + 1)\varepsilon$ is larger.

7.4 Few-shot fine-tuning with linear flow matching

Table 4 and Figure 9 summarise the few-shot study, and Figures 6–8 show one task (test task 8). Three observations follow.

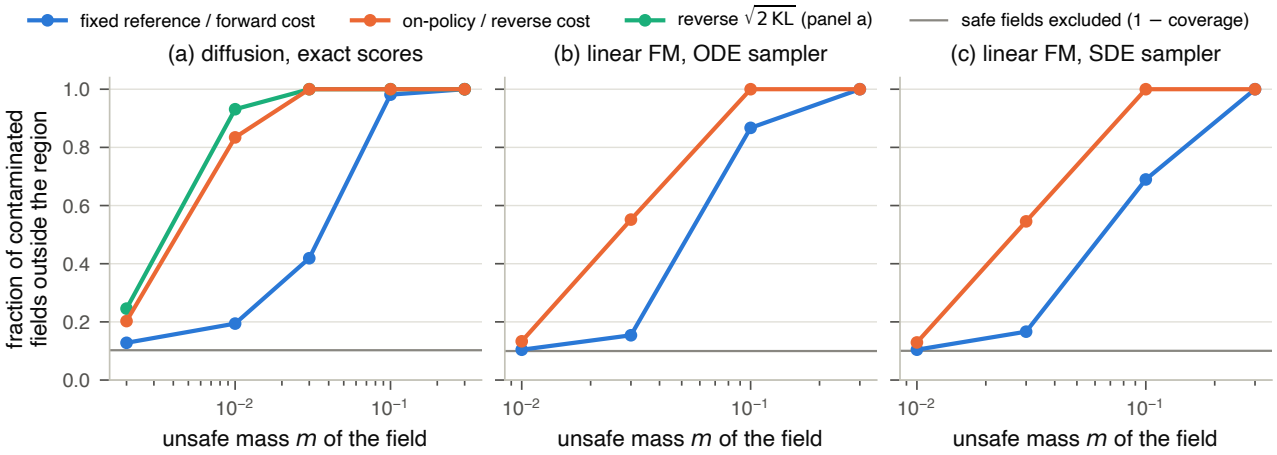


Figure 2: Fraction of contaminated fields outside the region as a function of their unsafe mass. The grey line is the fraction of held-out safe fields outside the region (one minus the coverage).

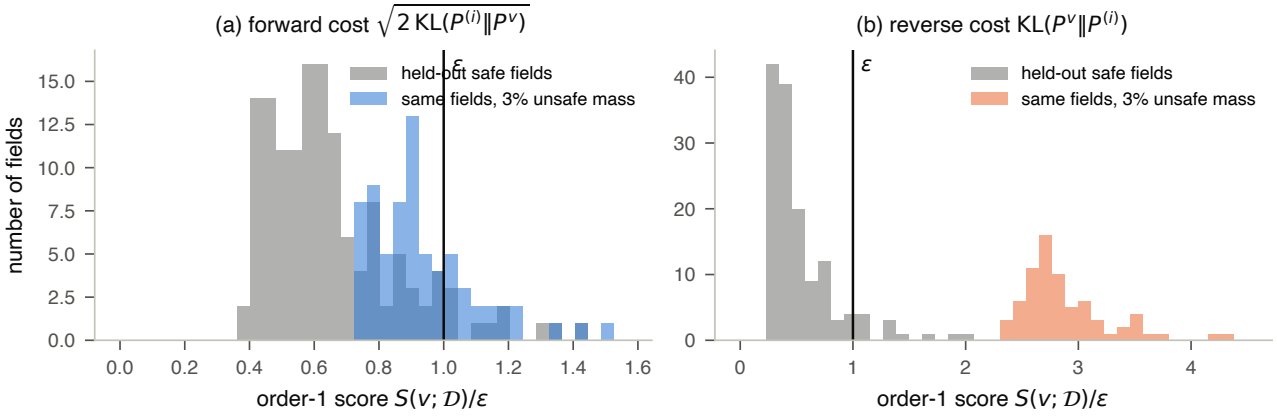


Figure 3: Order-1 scores, normalised by the conformal threshold, of the held-out safe fields and of the same fields with 3% unsafe mass (diffusion with exact scores, one split).

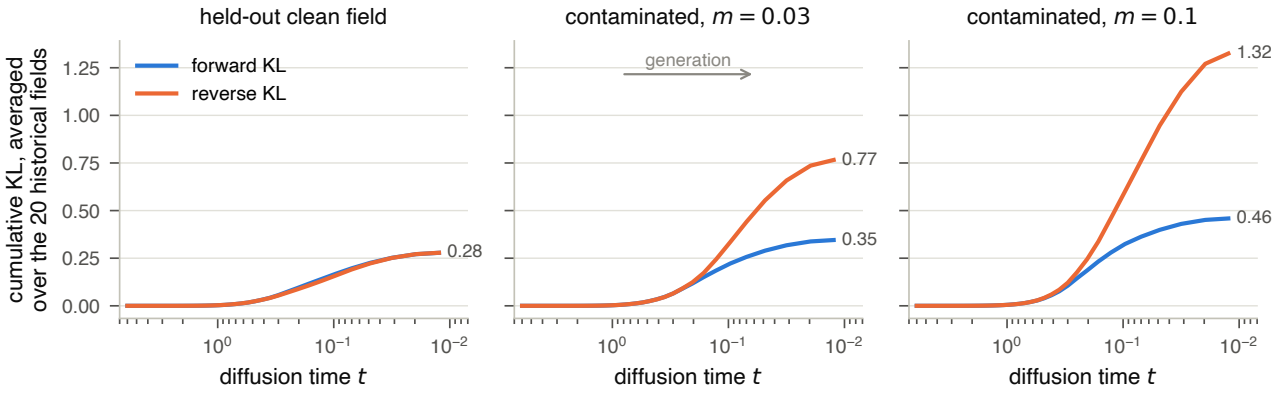


Figure 4: Cumulative path divergence between a held-out field and the 20 historical fields (averaged over the historical fields), as generation proceeds from $t = T$ to $t_{\min}$. Left: a clean field; middle and right: the same field with unsafe mass m. The final values are the averages that enter the order-1 score.

Table 3: Guaranteed bound on $P^v(A)$ for every field of the region (median over splits) and the largest actual $P^v(A)$ among the held-out and contaminated fields inside the region (median and maximum over splits). Violations: splits where the actual value exceeds the bound.

region and bound	bound (median)	actual (median)	actual (max)	violations
(a) diffusion, exact scores				
forward region, Theorem 1	0.283	0.0324	0.0947	0/200
reverse $\sqrt{2\mathrm{KL}}$ region, Theorem 1	0.284	0.0035	0.0137	0/200
reverse-KL region, Theorem 2	0.031	0.0123	0.0137	0/200
(b) linear flow matching, SDE sampler				
forward region, Theorem 1	0.553	0.0707	0.0737	0/200
reverse-KL region, Theorem 2	0.134	0.0145	0.0158	0/200

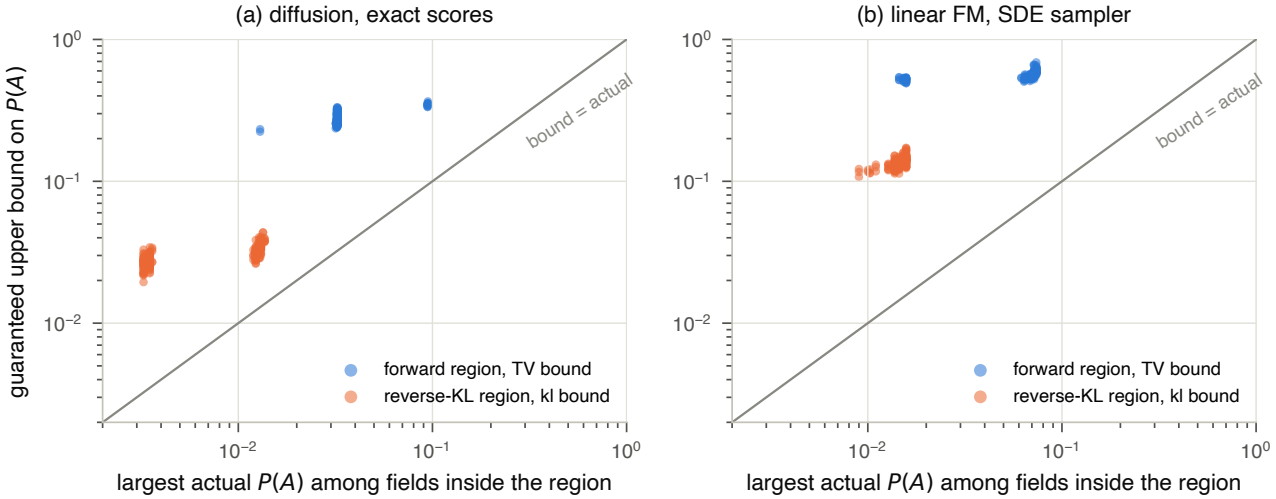


Figure 5: Each point is one random split: largest actual $P^v(A)$ among the fields inside the region against the guaranteed bound. Points above the diagonal are valid certificates.

Table 4: Few-shot fine-tuning on 32 held-out tasks (12 clean samples, and 12 clean plus 3 unsafe samples). MMD^2 is multiplied by 10^3. The last two columns give the fraction of fitted fields inside the fixed-Gram region and inside the on-policy region (contaminated case).

	clean data		contaminated data		inside region	
estimator	MMD^2	$P^v(A)$	MMD^2	$P^v(A)$	fixed G	on-policy
least squares, $\lambda = 10^{-5}$	168.2	0.0006	168.4	0.1663	0.00	0.00
least squares, $\lambda = 10^{-3}$	131.6	0.0037	144.0	0.1382	0.00	0.00
least squares, $\lambda = 10^{-1}$	154.8	0.0563	186.5	0.0969	0.00	0.00
shrinkage, $\gamma = 0.1$	135.0	0.0008	137.9	0.1476	0.00	0.00
shrinkage, $\gamma = 1$	43.2	0.0012	47.2	0.0686	0.00	0.00
shrinkage, $\gamma = 10$	10.8	0.0015	11.3	0.0087	1.00	1.00
conformal, fixed G (SOCP)	32.6	0.0013	29.6	0.0474	1.00	0.00
conformal, on-policy (Alg. 1)	40.2	0.0015	28.1	0.0246	0.00	1.00
centre θ_c (no data)	10.7	0.0014	10.7	0.0014	1.00	1.00
ample-data fit $f(\tau)$	5.1	0.0017	5.1	0.0017	0.91	0.94

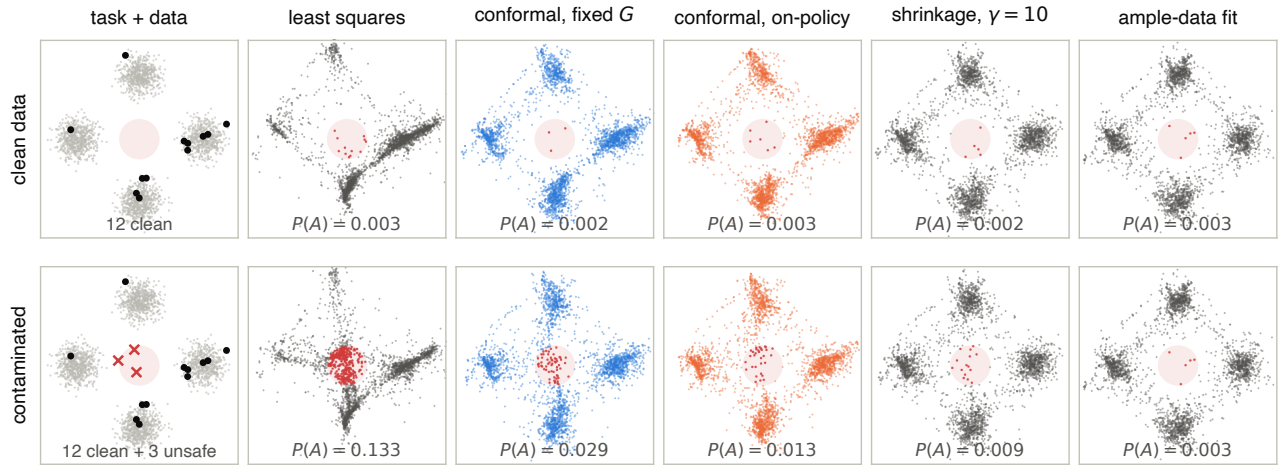


Figure 6: Generated samples for test task 8. First column: the clean task (grey) and the few-shot data (black, unsafe samples as red crosses). Other columns: ODE samples of each fitted field; samples in A are red. Top: clean data; bottom: contaminated data.

The constraint helps against least squares. The conformal SOCP beats the best least-squares fit on $32/32$ tasks with clean data and $32/32$ with contaminated data, with geometric-mean MMD^2 ratios 0.28 and 0.21; the on-policy variant gives 0.35 and 0.20. Each SOCP takes 0.31 s.

The constraint limits the unsafe mass, and the on-policy cost limits it more. With 20% contaminated samples, least squares puts 0.138 of its mass in A, the fixed-Gram constraint 0.047 and the on-policy constraint 0.025 (historical safe fits: 0.0017). The fixed-Gram solutions are inside the fixed-Gram region but never inside the on-policy region, i.e. the on-policy region detects their contamination. Algorithm 1 ended slightly infeasible in a fraction 0.70 of the runs, and the back-off then moved the solution by at most 0.011 of the way to the centre.

In this concentrated family, strong shrinkage towards the centre is better still. The centre alone, which uses no data, is within a factor of two of the oracle in MMD^2, and shrinkage with $\gamma = 10$ (tuned on the test tasks, an optimistic baseline) beats the conformal estimators in accuracy and in unsafe mass (0.0087). With 12 samples the least-squares objective pulls the constrained solution onto the boundary of the region, which is calibrated to contain 90% of the legitimate tasks and is therefore wide compared with the information in 12 samples. The neural network study of the manuscript (gauge versus projected gradient at $n \leq 5$) shows the same behaviour. The region is a safety and validity device, not an estimator by itself; combining it with shrinkage inside the region, $\min_{\boldsymbol{\theta}} \hat{L}(\boldsymbol{\theta}) + \gamma S(\boldsymbol{\theta}; \mathcal{D})$ subject to $S(\boldsymbol{\theta}; \mathcal{D}) \leq \varepsilon$, keeps the certificate and should recover the accuracy (not tested here).

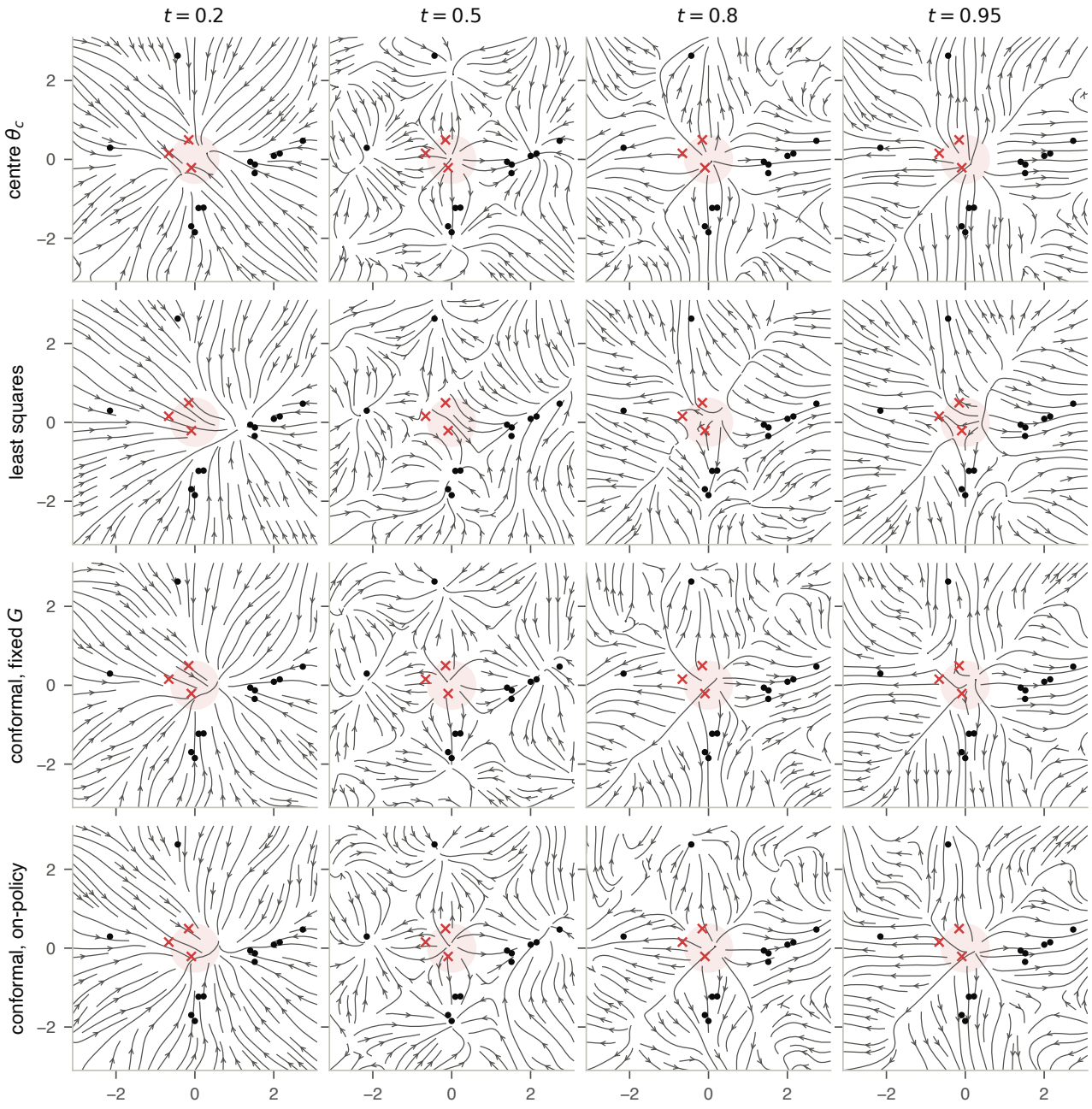


Figure 7: Streamlines of the time-dependent velocity fields $v_{\boldsymbol{\theta}}(\cdot, t)$ fitted on the contaminated data of test task 8, at four times. Early on ($t = 0.2$) every field contracts the noise towards the origin; the fields differ at late times. Least squares develops sinks at the unsafe samples inside A (red disk); the constrained fields route the flow to the ring.

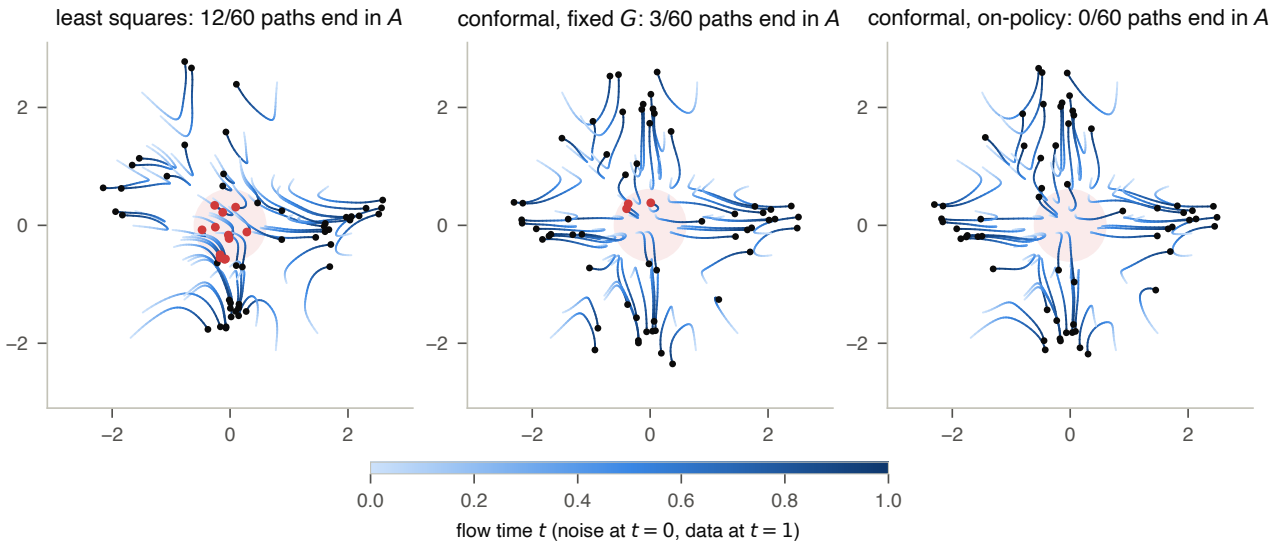


Figure 8: Sixty ODE trajectories of the fields of Figure 7, coloured by time; end points in A are red.

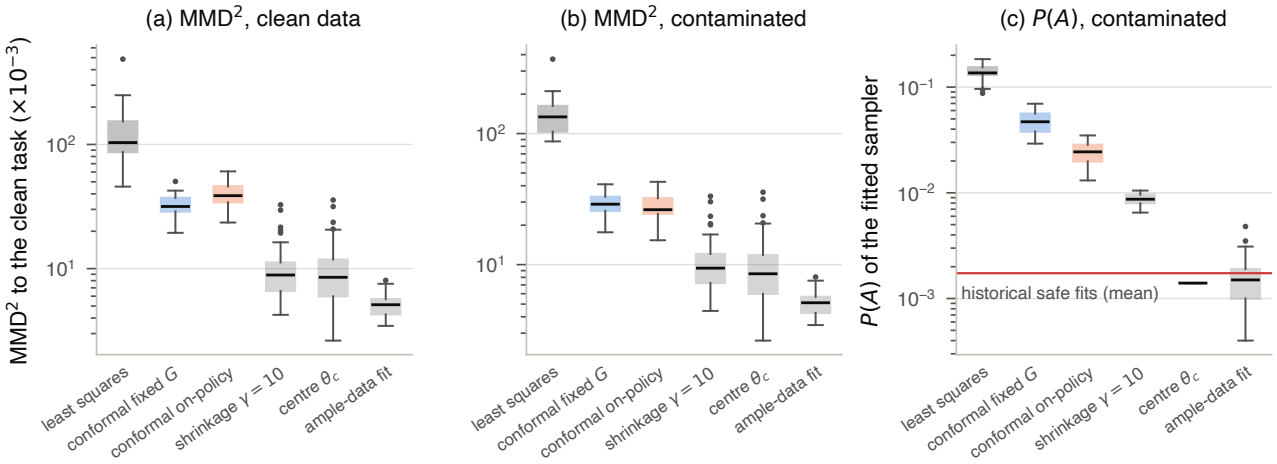


Figure 9: Few-shot fine-tuning over 32 held-out tasks: MMD^2 to the clean task with clean data (a) and with contaminated data (b), and probability of the unsafe set with contaminated data (c). The red line is the mean over the historical safe fits.

8 Prompt-conditioned flow matching: safe prompts induce safe fields

This section specialises the construction to the setting that motivates it: a trained text-to-image (or text-to-action) flow-matching model whose velocity field is shaped by the embedding of a prompt. The model is fixed and only the prompt varies, so a collection of safe prompts yields a collection of safe vector fields, and the order-1 region is built on these fields.

Setting. Let $v(\mathbf{x}, t; \mathbf{e})$ be a trained velocity field conditioned on a prompt embedding $\mathbf{e} \in \mathcal{E} \subseteq \mathbb{R}^p$ (for a tokenised prompt, the flattened output of a fixed text encoder). Each prompt defines a field $v_{\mathbf{e}} := v(\cdot, \cdot; \mathbf{e})$. The safe training prompts $\mathbf{e}_1, \dots, \mathbf{e}_M$ define the safe fields $v_{\mathbf{e}_1}, \dots, v_{\mathbf{e}_M}$, and the calibration prompts $\mathbf{e}_{M+1}, \dots, \mathbf{e}_{M+K}$ define the calibration fields; all are drawn from a distribution of approved prompts. A new prompt $\mathbf{e}$ may be unsafe, or adversarial. Let $\mu_0 := \mathcal{N}(0, I_n)$ and let $T_{\mathbf{e}} : \mathbb{R}^n \to \mathbb{R}^n$ map the initial noise $\mathbf{x}_0$ to the output of the deterministic sampler (the exact flow map, or the J-step Euler map). A fixed measurable map $\chi : \mathbb{R}^n \to \mathbb{R}^q$ (the identity, a decoder, an image encoder, or the features of a safety classifier) chooses the space in which outputs are compared. A cost on fields is pulled back to prompts, $c(\mathbf{e}, \mathbf{e}') := c(v_{\mathbf{e}}, v_{\mathbf{e}'})$, and the order-1 score of a prompt is $S_c^w(\mathbf{e}; \mathcal{D}) = \frac{1}{M+1} \sum_i w_i \, c(\mathbf{e}, \mathbf{e}_i)$ as in Definition 1.

8.1 Straight flows: when time collapses

Definition 5 (Straight flow, one-step map, straightness defect)**.** *The field $v_{\mathbf{e}}$ is* straight *if every trajectory of $\dot{X}_t = v_{\mathbf{e}}(X_t, t)$ is a segment traversed at constant speed: $X_t = X_0 + t\,(X_1 - X_0)$ for all $t \in [0,1]$. The* one-step map *is $\tilde{T}_{\mathbf{e}}(\mathbf{x}_0) \coloneqq \mathbf{x}_0 + v_{\mathbf{e}}(\mathbf{x}_0, 0)$ and the* straightness defect *is $\zeta_{\mathbf{e}} \coloneqq \|T_{\mathbf{e}} - \tilde{T}_{\mathbf{e}}\|_{L^2(\mu_0)}$, where $\|F\|_{L^2(\mu_0)} \coloneqq (\mathbb{E}_{\mathbf{x}_0 \sim \mu_0} \|F(\mathbf{x}_0)\|^2)^{1/2}$.*

Lemma 7 (Straight flows)**.** *Let $v_{\mathbf{e}}$ and $v_{\mathbf{e}'}$ be straight, let $\mathbf{x}_0 \in \mathbb{R}^n$, and let $X_t^{\mathbf{e}}$ and $X_t^{\mathbf{e}'}$ be their trajectories from $\mathbf{x}_0$. Then: (i) $v_{\mathbf{e}}(X_t^{\mathbf{e}}, t) = v_{\mathbf{e}}(\mathbf{x}_0, 0)$ for all t: along each trajectory the velocity is constant, in direction and in magnitude; (ii) $T_{\mathbf{e}} = \tilde{T}_{\mathbf{e}}$, i.e. $\zeta_{\mathbf{e}} = 0$, and $X_t^{\mathbf{e}} = \mathbf{x}_0 + t\, v_{\mathbf{e}}(\mathbf{x}_0, 0)$; (iii) $\sup_{t\in[0,1]} \|X_t^{\mathbf{e}} - X_t^{\mathbf{e}'}\| = \|v_{\mathbf{e}}(\mathbf{x}_0, 0) - v_{\mathbf{e}'}(\mathbf{x}_0, 0)\|$ and $\int_0^1 \|v_{\mathbf{e}}(X_t^{\mathbf{e}}, t) - v_{\mathbf{e}'}(X_t^{\mathbf{e}'}, t)\|^2\, dt = \|v_{\mathbf{e}}(\mathbf{x}_0, 0) - v_{\mathbf{e}'}(\mathbf{x}_0, 0)\|^2$; (iv) at a fixed point $\mathbf{x}$, $t \mapsto v_{\mathbf{e}}(\mathbf{x}, t)$ need not be constant, in direction or in magnitude. For instance, the field with $v(\mathbf{x}, 0) = A\mathbf{x}$ and $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is straight, $v(\mathbf{x}, t) = A(I + tA)^{-1}\mathbf{x}$, and at $\mathbf{x} = (1, 0)$ its direction rotates from 90° at $t = 0$ to 45° at $t = 1$.*

Proof: (i) By Definition 5, $X_t^{\mathbf{e}} = \mathbf{x}_0 + t(X_1^{\mathbf{e}} - \mathbf{x}_0)$, so $\dot{X}_t^{\mathbf{e}} = X_1^{\mathbf{e}} - \mathbf{x}_0$ does not depend on t. Since $\dot{X}_t^{\mathbf{e}} = v_{\mathbf{e}}(X_t^{\mathbf{e}}, t)$, the velocity along the trajectory equals its value at $t = 0$, which is $v_{\mathbf{e}}(\mathbf{x}_0, 0)$. (ii) By (i), $X_1^{\mathbf{e}} - \mathbf{x}_0 = v_{\mathbf{e}}(\mathbf{x}_0, 0)$, i.e. $T_{\mathbf{e}}(\mathbf{x}_0) = \tilde{T}_{\mathbf{e}}(\mathbf{x}_0)$; the trajectory then follows from Definition 5. (iii) By (ii), $X_t^{\mathbf{e}} - X_t^{\mathbf{e}'} = t\,(v_{\mathbf{e}}(\mathbf{x}_0, 0) - v_{\mathbf{e}'}(\mathbf{x}_0, 0))$, whose norm is largest at $t = 1$. By (i), the integrand of the second identity does not depend on t and equals $\|v_{\mathbf{e}}(\mathbf{x}_0, 0) - v_{\mathbf{e}'}(\mathbf{x}_0, 0)\|^2$. (iv) The trajectories of the example are $X_t = (I + tA)\mathbf{x}_0$, segments traversed at the constant speed $A\mathbf{x}_0$, so the field is straight. The matrix $I + tA$ is invertible (its eigenvalues are $1 \pm it$), so the point $\mathbf{x}$ is reached at time t from $\mathbf{x}_0 = (I + tA)^{-1}\mathbf{x}$, and $v(\mathbf{x}, t) = A\mathbf{x}_0 = A(I + tA)^{-1}\mathbf{x}$. With $(I + tA)^{-1} = \frac{1}{1+t^2}\left(\begin{smallmatrix} 1 & t \\ -t & 1 \end{smallmatrix}\right)$ one gets $v((1, 0), t) = \frac{1}{1+t^2}(t, 1)$, whose angle with the first axis is $\arctan(1/t)$. □

Lemma 7 makes precise the intuition that in a straight ("linear") flow the field does not change angle through time: this holds *along trajectories*, not at a fixed point. Its practical content is (ii)–(iii). For a straight flow the whole time-dependent field is summarised by its slice at $t = 0$, the sampler is a single step, and every time-integrated comparison of two fields along their own trajectories reduces to a comparison of these slices.

Remark 5 (The linear interpolant does not make the learned flow straight)**.** *Training with the linear interpolant on independent noise/data pairs makes the* conditional *paths straight, not the learned marginal flow. Already for Gaussian data $\mathcal{N}(0, \Sigma)$, with $C_t \coloneqq (1-t)^2 I + t^2\Sigma$, the marginal velocity is $v^\star(\mathbf{x}, t) = (t\Sigma - (1-t)I)\, C_t^{-1}\mathbf{x}$ (Gaussian conditioning, since $\mathrm{Cov}(\mathbf{x}_1 - \mathbf{x}_0, \mathbf{x}_t) = t\Sigma - (1-t)I$), and its flow is $X_t = C_t^{1/2} X_0$: indeed $\frac{d}{dt} C_t^{1/2} = \frac{1}{2} C_t^{-1/2} \dot{C}_t = (t\Sigma - (1-t)I)\, C_t^{-1/2}$, because all the matrices involved commute. In an eigenbasis of Σ the j-th coordinate is scaled by $\sqrt{(1-t)^2 + t^2\lambda_j}$, which is not affine in t, so the trajectories are not segments; when the eigenvalues differ, the direction of X_t changes as well. Reflow (Liu et al. 2023) straightens the coupling; one-step text-to-image generators are obtained by reflow and distillation (Liu et al. 2024); text-to-image models trained with the rectified-flow objective (Esser et al. 2024) are not reflowed, so their flows need not be straight. The defect $\zeta_{\mathbf{e}}$ measures the deviation, and it enters the bounds below.*

8.2 Map metrics and their certificate

Definition 6 (Map metrics)**.** *For prompts $\mathbf{e}, \mathbf{e}'$ let*

$$d_T(\mathbf{e}, \mathbf{e}') \coloneqq \|T_{\mathbf{e}} - T_{\mathbf{e}'}\|_{L^2(\mu_0)}, \qquad d_0(\mathbf{e}, \mathbf{e}') \coloneqq \|\tilde{T}_{\mathbf{e}} - \tilde{T}_{\mathbf{e}'}\|_{L^2(\mu_0)} = \|v_{\mathbf{e}}(\cdot, 0) - v_{\mathbf{e}'}(\cdot, 0)\|_{L^2(\mu_0)},$$
$$d_\chi(\mathbf{e}, \mathbf{e}') \coloneqq \|\chi \circ T_{\mathbf{e}} - \chi \circ T_{\mathbf{e}'}\|_{L^2(\mu_0)}.$$

Lemma 8 (Map metrics are pseudometrics)**.** *d_T, d_0 and d_χ are pseudometrics on prompts (equivalently, on fields), and $|d_T(\mathbf{e}, \mathbf{e}') - d_0(\mathbf{e}, \mathbf{e}')| \le \zeta_{\mathbf{e}} + \zeta_{\mathbf{e}'}$. In particular $d_T = d_0$ on straight fields.*

Proof: Each is the $L^2(\mu_0)$ distance between two functions of $\mathbf{x}_0$; it is therefore symmetric, zero on the diagonal, and satisfies the triangle inequality of the $L^2(\mu_0)$ norm. For the bound, the triangle inequality in $L^2(\mu_0)$ gives $\|T_{\mathbf{e}} - T_{\mathbf{e}'}\| \le \|T_{\mathbf{e}} - \tilde{T}_{\mathbf{e}}\| + \|\tilde{T}_{\mathbf{e}} - \tilde{T}_{\mathbf{e}'}\| + \|\tilde{T}_{\mathbf{e}'} - T_{\mathbf{e}'}\|$, i.e. $d_T \le \zeta_{\mathbf{e}} + d_0 + \zeta_{\mathbf{e}'}$, and symmetrically $d_0 \le \zeta_{\mathbf{e}} + d_T + \zeta_{\mathbf{e}'}$. By Lemma 7(ii), $\zeta = 0$ on straight fields. □

By Lemma 1, the order-1 score with any of these costs is exactly the order-1 Wasserstein distance between consecutive empirical distributions whose samples are the safe fields, with the metric defined on the space of fields. In practice the outputs $\chi(T_{\mathbf{e}_i}(\mathbf{x}_0^{(l)}))$ of the safe prompts on a fixed set of probe noises $\mathbf{x}_0^{(1)}, \dots, \mathbf{x}_0^{(L)}$ are computed once and stored. Scoring a new prompt then costs one run of the sampler on the probe noises (one network evaluation per probe for d_0), plus the encoder χ. The memory is $M \times L \times q$ numbers, which favours a low-dimensional χ.

Theorem 3 (Output certificate for map metrics)**.** *Let $d = d_\chi$ (with $\chi = \mathrm{id}$, $d = d_T$), let w be a probability vector and $\varepsilon \ge 0$, and assume $S_d^w(\mathbf{e}; \mathcal{D}) \le \varepsilon$. Let $\mathbf{x}_0 \sim \mu_0$ and $I \sim w$ be independent, $Y \coloneqq \chi(T_{\mathbf{e}}(\mathbf{x}_0))$ and $Y' \coloneqq \chi(T_{\mathbf{e}_I}(\mathbf{x}_0))$. Then: (i) Y has the law of the (encoded) outputs of the prompt $\mathbf{e}$, Y' has the w-mixture of the laws of the outputs of the safe prompts, and $\mathbb{E}\,\|Y - Y'\| \le (M+1)\varepsilon$; hence $W_1(\mathrm{law}(Y), \mathrm{law}(Y')) \le (M+1)\varepsilon$; (ii) for every Borel set $B \subseteq \mathbb{R}^q$ and every $r > 0$, with $B^{+r} \coloneqq \{\mathbf{y} : \mathrm{dist}(\mathbf{y}, B) < r\}$,*

$$\mathbb{P}\big(\chi(T_{\mathbf{e}}(\mathbf{x}_0)) \in B\big) \;\le\; \sum_i w_i\, \mathbb{P}\big(\chi(T_{\mathbf{e}_i}(\mathbf{x}_0)) \in B^{+r}\big) + \frac{(M+1)\varepsilon}{r};$$

(iii) if the region is defined with d_0 instead (i.e. $S^w_{d_0}(\mathbf{e};\mathcal{D}) \le \varepsilon$) and $\chi = \mathrm{id}$, then (i)–(ii) hold for the outputs of the full sampler with $(M+1)\varepsilon$ replaced by $(M+1)\varepsilon + \zeta_{\mathbf{e}} + \sum_i w_i \zeta_{\mathbf{e}_i}$.

Proof: (i) The marginal statements follow from the independence of $\mathbf{x}_0$ and I. Conditioning on I and using $\mathbb{E}\,\|Z\| \le (\mathbb{E}\,\|Z\|^2)^{1/2}$, $\mathbb{E}\,\|Y - Y'\| = \sum_i w_i\, \mathbb{E}\,\|\chi(T_{\mathbf{e}}(\mathbf{x}_0)) - \chi(T_{\mathbf{e}_i}(\mathbf{x}_0))\| \le \sum_i w_i\, d_\chi(\mathbf{e},\mathbf{e}_i)$, which is at most $(M+1)\varepsilon$ by (3). Since (Y, Y') is a coupling of the two laws, W_1 is at most $\mathbb{E}\,\|Y - Y'\|$. (ii) As in Step 4 of Proposition 3: if $Y \in B$ and $\|Y - Y'\| < r$, then $Y' \in B^{+r}$, so $\mathbb{P}(Y \in B) \le \mathbb{P}(Y' \in B^{+r}) + \mathbb{P}(\|Y - Y'\| \ge r)$; Markov's inequality and (i) bound the last term by $(M+1)\varepsilon/r$. (iii) By Lemma 8, $d_T(\mathbf{e},\mathbf{e}_i) \le d_0(\mathbf{e},\mathbf{e}_i) + \zeta_{\mathbf{e}} + \zeta_{\mathbf{e}_i}$; multiplying by w_i and summing, $\sum_i w_i d_T(\mathbf{e},\mathbf{e}_i) \le (M+1)\varepsilon + \zeta_{\mathbf{e}} + \sum_i w_i \zeta_{\mathbf{e}_i}$. Repeat (i)–(ii) with this bound. □

Theorem 3 is Proposition 3 with the Grönwall factor replaced by 1: comparing outputs under shared noise, instead of fields along trajectories, removes the propagation of errors that made Proposition 3 vacuous. The certificate holds deterministically for every prompt whose score is at most ε, safe or not, and for every set B at once, so the method never needs to know B. Its strength depends on two things: the budget $(M+1)\varepsilon$, which the floor of Remark 4 ties to the spread of the safe prompts, and the space in which outputs are compared, which χ chooses. A pixel or latent distance is not semantic; a fixed image encoder makes the certificate semantic in its feature space.

Proposition 7 (Sensitivity of map metrics to rare unsafe outputs)**.** *Let $\Delta_{\mathrm{s}} > 0$ and let $N \subseteq \mathbb{R}^n$ be a set of noises with $\mu_0(N) = m$ such that $\|\chi(T_{\mathbf{e}}(\mathbf{x})) - \chi(T_{\mathbf{e}_i}(\mathbf{x}))\| \ge \Delta_{\mathrm{s}}$ for all $\mathbf{x} \in N$ and all $i \in [\![M]\!]$. Then $d_\chi(\mathbf{e},\mathbf{e}_i) \ge \sqrt{m}\,\Delta_{\mathrm{s}}$ for every i, hence $S_{d_\chi}(\mathbf{e};\mathcal{D}) \ge \sqrt{m}\,\Delta_{\mathrm{s}}/(M+1)$, and $\mathbf{e}$ lies outside the region whenever $\sqrt{m}\,\Delta_{\mathrm{s}} > (M+1)\varepsilon$.*

Proof: $d_\chi(\mathbf{e},\mathbf{e}_i)^2 = \int \|\chi T_{\mathbf{e}} - \chi T_{\mathbf{e}_i}\|^2 d\mu_0 \ge \int_N \|\chi T_{\mathbf{e}} - \chi T_{\mathbf{e}_i}\|^2 d\mu_0 \ge m\Delta_{\mathrm{s}}^2$. Averaging over i with weights $1/M$ and dividing by $M+1$ gives the bound on the score. □

A prompt that sends a fraction m of the noise to outputs at distance Δ_{s} from the safe outputs therefore costs at least $\sqrt{m}\,\Delta_{\mathrm{s}}$ under a map metric, compared with at most $\log \frac{1}{1-m} \approx m$ under a forward divergence (Proposition 2). Map metrics detect rare unsafe outputs when these are far from the safe outputs in the space chosen by χ. They do not use the rarity of the unsafe region; only the reverse divergence of Theorem 2 does.

8.3 Forcing a prompt into the safe set

A prompt whose score exceeds ε can be refused, or its field can be moved into the region. Two mechanisms have closed forms.

Proposition 8 (One-step blending)**.** *Let $\mathbf{e}_c$ be a prompt with $S_{d_0}(\mathbf{e}_c;\mathcal{D}) < \varepsilon < S_{d_0}(\mathbf{e};\mathcal{D})$ (for instance the safe training prompt nearest to $\mathbf{e}$ among those whose in-sample score is below ε). For $\lambda \in [0,1]$ let $\tilde{T}_\lambda := (1-\lambda)\tilde{T}_{\mathbf{e}} + \lambda \tilde{T}_{\mathbf{e}_c}$, that is, $\tilde{T}_\lambda(\mathbf{x}_0) = \mathbf{x}_0 + (1-\lambda)v(\mathbf{x}_0,0;\mathbf{e}) + \lambda v(\mathbf{x}_0,0;\mathbf{e}_c)$, and $S(\lambda) := \frac{1}{M(M+1)} \sum_i \|\tilde{T}_\lambda - \tilde{T}_{\mathbf{e}_i}\|_{L^2(\mu_0)}$. Then S is convex on $[0,1]$, and $\lambda^\star := \frac{S_{d_0}(\mathbf{e};\mathcal{D}) - \varepsilon}{S_{d_0}(\mathbf{e};\mathcal{D}) - S_{d_0}(\mathbf{e}_c;\mathcal{D})} \in (0,1)$ satisfies $S(\lambda^\star) \le \varepsilon$. Consequently the one-step outputs $\tilde{T}_{\lambda^\star}(\mathbf{x}_0)$ satisfy Theorem 3(i)–(ii) with respect to the one-step outputs of the safe prompts.*

Proof: $\tilde{T}_\lambda - \tilde{T}_{\mathbf{e}_i} = (1-\lambda)(\tilde{T}_{\mathbf{e}} - \tilde{T}_{\mathbf{e}_i}) + \lambda(\tilde{T}_{\mathbf{e}_c} - \tilde{T}_{\mathbf{e}_i})$ is affine in λ, so each norm, and hence their sum, is convex in λ. Since $S(0) = S_{d_0}(\mathbf{e};\mathcal{D})$ and $S(1) = S_{d_0}(\mathbf{e}_c;\mathcal{D})$, Proposition 6 (convex case) gives $S(\lambda^\star) \le \varepsilon$. The map $\tilde{T}_{\lambda^\star}$ plays the role of $T_{\mathbf{e}}$ in Theorem 3 with $\chi = \mathrm{id}$, and the proof of the theorem applies verbatim. □

The blend costs two network evaluations, as classifier-free guidance does (Ho and Salimans 2022), and for a straight model it is the whole sampler. With a multi-step sampler, blending the velocities at every step does not blend the maps, because the flow is nonlinear. The bisection of Proposition 6, run on d_T with the blended sampler re-simulated on the probe noises, nevertheless returns a field inside the region, since its invariant needs no convexity. The choice of $\mathbf{e}_c$ trades fidelity against cost. The nearest feasible safe prompt keeps as much of the request as the region allows. The geometric median of the safe maps, which is a convex combination $\sum_j \alpha_j \tilde{T}_{\mathbf{e}_j}$ of them because $L^2(\mu_0)$ is a Hilbert space, requires M network evaluations per step.

Proposition 9 (Prompt projection for a model affine in the prompt)**.** *Assume $v(\mathbf{x},0;\mathbf{e}) = \mathbf{a}(\mathbf{x}) + J(\mathbf{x})\mathbf{e}$ for all $\mathbf{e}$, with $J(\mathbf{x}) \in \mathbb{R}^{n\times p}$ and $\mathbb{E}_{\mu_0}\,\|J\|_F^2 < \infty$; let $G_J := \mathbb{E}_{\mu_0}[J^\top J]$ and assume $G_J \succ 0$. Then: (i) $d_0(\mathbf{e},\mathbf{e}') = \|\mathbf{e} - \mathbf{e}'\|_{G_J}$, the pull-back of the field metric to prompt embeddings; (ii) if $S_{d_0}(\mathbf{e}_c;\mathcal{D}) < \varepsilon$ for some $\mathbf{e}_c$ and $R := M(M+1)\varepsilon$, the projection*

$$\mathbf{e}^\star := \operatorname*{arg\,min}_{\mathbf{e}\in\mathbb{R}^p} \|\mathbf{e} - \mathbf{e}_{\mathrm{new}}\|_{G_J}^2 \quad \textit{s.t.} \quad \sum_{i=1}^{M} \|\mathbf{e} - \mathbf{e}_i\|_{G_J} \le R$$

exists, is unique, solves a second-order cone program, and its field lies in the region; (iii) $\mathbf{e}^\star$ belongs to the convex hull of $\{\mathbf{e}_{\mathrm{new}}, \mathbf{e}_1, \dots, \mathbf{e}_M\}$, so it can be computed over the simplex of $\mathbb{R}^{M+1}$ from the Gram matrix $K_{ab} := \mathbf{e}_a^\top G_J \mathbf{e}_b$, $a,b \in \{0,\dots,M\}$, $\mathbf{e}_0 := \mathbf{e}_{\mathrm{new}}$, whatever the embedding dimension p.

Proof: (i) $v(\cdot,0;\mathbf{e}) - v(\cdot,0;\mathbf{e}') = J(\mathbf{e} - \mathbf{e}')$; take the squared norm and the expectation, as in Proposition 4(a). (ii) The feasible set is closed, convex and non-empty (it contains $\mathbf{e}_c$), and the objective is strictly convex and coercive because $G_J \succ 0$, so the minimiser exists and is unique; the second-order cone reformulation is that of Proposition 5. By (i),

the constraint reads $S_{d_0}(\mathbf{e}^\star;\mathcal{D}) \le \varepsilon$. (iii) If the constraint is inactive, $\mathbf{e}^\star = \mathbf{e}_{\text{new}}$; if $\mathbf{e}^\star = \mathbf{e}_j$ for some j, the claim is immediate. Otherwise the constraint function is differentiable at $\mathbf{e}^\star$, Slater's condition holds because $\mathbf{e}_c$ is strictly feasible, and the Karush–Kuhn–Tucker conditions give a multiplier $\nu \ge 0$ with $2G_J(\mathbf{e}^\star - \mathbf{e}_{\text{new}}) + \nu \sum_i G_J(\mathbf{e}^\star - \mathbf{e}_i)/r_i = 0$, where $r_i := \|\mathbf{e}^\star - \mathbf{e}_i\|_{G_J} > 0$. Multiplying by G_J^{-1} and solving, $\mathbf{e}^\star = \big(2\mathbf{e}_{\text{new}} + \nu \sum_i \mathbf{e}_i/r_i\big)/\big(2 + \nu \sum_i 1/r_i\big)$, a convex combination. Writing $\mathbf{e} = \sum_a \alpha_a \mathbf{e}_a$, every norm in the problem is $\|\mathbf{e} - \mathbf{e}_b\|_{G_J}^2 = (\boldsymbol{\alpha} - \mathbf{1}_b)^\top K (\boldsymbol{\alpha} - \mathbf{1}_b)$, with $\mathbf{1}_b$ the b-th unit vector. □

For a model that is not affine in the prompt, J is the Jacobian of $v(\mathbf{x}, 0; \cdot)$ at $\mathbf{e}_{\text{new}}$, the projection is exact to first order only, and the score of the projected prompt must be checked on the true model, with the back-off of Proposition 6 if it exceeds ε. The entries of K require only Jacobian–vector products $J(\mathbf{x})\mathbf{e}_a$ on the probe noises, $M + 1$ per probe.

8.4 Local safe sets for diverse prompts

When the safe prompts are diverse, the floor of Remark 4 makes the uniform region wide: its budget is at least half the mean distance between, say, the fields of a prompt about a cat and of a prompt about a car, and forcing then pulls towards an "average" safe output. Two localisations keep the guarantees.

Prompt-adaptive weights. Let the weights depend on the new prompt, e.g. $w_i(\mathbf{e}) \propto \exp(-\|\mathbf{e} - \mathbf{e}_i\|^2/(2h^2))$, or uniform on the k safe prompts nearest to $\mathbf{e}$ in embedding space, as in the task-adaptive score $g_{d,1}$ of the manuscript. The map $\mathbf{e} \mapsto S_d^{w(\mathbf{e})}(\mathbf{e};\mathcal{D})$ is still a fixed function of the prompt once the training prompts are fixed. Proposition 1 therefore applies when every calibration prompt is scored with its own weights, and Theorem 3 holds with the $w(\mathbf{e})$-mixture, i.e. relative to the safe prompts most similar to the request.

Order-∞. The manuscript proves (`thm:regions` of its region-equivalence section, valid for any symmetric d) that the order-∞ region without cuts is the union of the closed d-balls of radius ε_∞ around the safe fields whenever ε_∞ is at least the bottleneck of the minimum spanning tree of the safe fields, i.e. whenever their threshold graph at level ε_∞ is connected. If $d_\chi(\mathbf{e}, \mathbf{e}_j) \le \varepsilon_\infty$ for some j, the proof of Theorem 3 with w the Dirac mass at j gives $\mathbb{P}(\chi(T_{\mathbf{e}}(\mathbf{x}_0)) \in B) \le \max_j \mathbb{P}(\chi(T_{\mathbf{e}_j}(\mathbf{x}_0)) \in B^{+r}) + \varepsilon_\infty/r$: the request is certified relative to *one* safe prompt, not to the average of all of them.

9 A catalogue of design choices

The construction has four independent components: the cost between fields, the shape of the region, the enforcement mechanism and the calibration procedure. This section lists the options that the analysis supports, with the guarantee each one carries.

9.1 Costs between fields

All certificates of this note follow from one principle.

Lemma 9 (Convexity principle)**.** *Let $\mathfrak{D}$ be a $[0, +\infty]$-valued function of pairs of probability measures that is convex in its second argument: $\mathfrak{D}(P, \sum_i w_i Q_i) \le \sum_i w_i \mathfrak{D}(P, Q_i)$. If $c(v, v_i) := \mathfrak{D}(P^v, P^{(i)})$ and $S_c^w(v;\mathcal{D}) \le \varepsilon$, then $\mathfrak{D}(P^v, \bar{P}_w) \le (M+1)\varepsilon$.*

Proof: $\mathfrak{D}(P^v, \bar{P}_w) \le \sum_i w_i \mathfrak{D}(P^v, P^{(i)}) = (M+1) S_c^w(v;\mathcal{D}) \le (M+1)\varepsilon$, by convexity and (3). □

The hypothesis holds for: the total variation and the reverse divergence $\mathrm{KL}(P\|\cdot)$ (Step 1 of Theorems 1 and 2); the forward divergence $\mathfrak{D}(P, Q) := \mathrm{KL}(Q\|P)$ (joint convexity of the relative entropy); W_1 (a mixture of couplings of the pairs (P, Q_i) is a coupling of $(P, \sum_i w_i Q_i)$); the maximum mean discrepancy $\mathrm{MMD}_k(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}_k}$, a norm of a difference of mean embeddings (Gretton et al. 2012); and $|\mathbb{E}_P\, g - \mathbb{E}_Q\, g|$ for a fixed function g. What the certificate means depends on $\mathfrak{D}$:

- a forward divergence certifies that the candidate *covers* the safe behaviour ($\mathrm{KL}(\bar{P}_w\|P^v)$ small);
- a reverse divergence certifies that it *avoids* what the safe samplers avoid;
- W_1 certifies that its outputs are close to theirs.

Table 5 lists the costs.

Three remarks on the table. First, the costs differ in what they can see: the forward divergence is blind to small unsafe modes, the map metrics see unsafe outputs that are far from the safe ones in the chosen space, and the reverse divergence sees unsafe outputs that are rare under the safe samplers, wherever they are. Second, the law distances compare distributions rather than maps, so they do not need shared noise and remain meaningful across samplers, but they are weaker than the map metrics: W_1 between the laws is at most the corresponding map distance. Third, the scalar functional is the degenerate end of the spectrum: with g a safety classifier, the region reduces to a conformal test on the expected classifier score, which is fully semantic but sees nothing else. The angular metric discards magnitudes, which decide how far samples move, so it is not suitable alone.

Table 5: Costs between fields. "Metric": the cost is a pseudometric, so Lemma 1 gives the order-1 Wasserstein interpretation. "Convex": in the field (or in the parameters of a linear model). Last column: what the cost charges for a fraction m of outputs in a region of safe probability q (with $p \geq m$ the candidate's probability of that region), at distance Δ_{s} from the safe outputs.

cost	definition	metric	convex	certificate	unsafe mass m
fixed-reference L^2	$\|v - v'\|_{L^2(\rho)}$, ρ the pooled safe path (Prop. 4)	yes	yes	none directly	weak
forward KL	$\mathrm{KL}(P^{(i)} \| P^v)$, historical paths (Cor. 1)	no	yes	TV (Thm. 1); covers safe behaviour	$\leq \log \frac{1}{1-m}$
reverse KL	$\mathrm{KL}(P^v \| P^{(i)})$, own paths (Cor. 1)	no	no	rare events (Thm. 2)	$\geq p \log \frac{p}{eq}$
output map	$d_T = \|T_v - T_{v'}\|_{L^2(\mu_0)}$	yes	no	W_1, events (Thm. 3)	$\geq \sqrt{m}\,\Delta_{\mathrm{s}}$
one-step map	$d_0 = \|v(\cdot,0) - v'(\cdot,0)\|_{L^2(\mu_0)}$	yes	yes	as above, plus defects	$\geq \sqrt{m}\,\Delta_{\mathrm{s}} - \zeta - \zeta'$
feature map	$d_\chi = \|\chi T_v - \chi T_{v'}\|_{L^2(\mu_0)}$	yes	no	W_1 in the space of χ	$\geq \sqrt{m}\,\Delta_{\mathrm{s}}$ there
law distance	W_1 or MMD_k between the laws of the encoded outputs	yes	no	Lemma 9	about $m\Delta_{\mathrm{s}}$ (W_1)
scalar functional	$\lvert \mathbb{E}_{P^v} g - \mathbb{E}_{P^{(i)}} g \rvert$, g e.g. a safety classifier	yes	no	on $\mathbb{E}\, g$ only	$m\times$ score gap
pull-back metric	$\|\mathbf{e} - \mathbf{e}'\|_{G_J}$ (Prop. 9)	yes	yes (in $\mathbf{e}$)	as d_0	as d_0, to first order
angular	$\arccos \frac{\langle v, v' \rangle_\rho}{\|v\|_\rho \|v'\|_\rho}$	on directions	no	none	ignores magnitudes

9.2 Regions

- *Uniform order*-1 ($w_i = 1/M$): the region of the manuscript; certificates relative to the average of the safe prompts.
- *Prompt-adaptive order*-1 (Section 8.4): kernel or k-nearest-neighbour weights in embedding space; certificates relative to the safe prompts closest to the request.
- *Order*-∞: the union of balls around the safe fields, or the bottleneck region with n cuts for safe prompts that form $n + 1$ clusters; certificates relative to one safe prompt.
- *Cluster-wise regions*: one region per concept cluster of safe prompts, each calibrated on its own cluster (Mondrian conformal prediction, Vovk et al. 2005), and the request is assigned to its nearest cluster.
- *Intersections*: $\{S_{c_1} \leq \varepsilon_1\} \cap \{S_{c_2} \leq \varepsilon_2\}$, e.g. a map metric for geometry and the reverse divergence for rare events; with thresholds calibrated at levels β_1 and β_2 the coverage is at least $1 - \beta_1 - \beta_2$ (union bound), and both certificates hold.
- *Contrastive scores*: when unsafe prompts $\mathbf{u}_1, \ldots, \mathbf{u}_L$ are also available, a score such as $S_c(\mathbf{e}; \mathcal{D}) - \kappa\, S_c(\mathbf{e}; \{\mathbf{u}_l\})$ keeps the coverage of Proposition 1 (any fixed score does) and can increase the power. It loses the certificate unless it is intersected with a safe-only region.

9.3 Enforcement

- *Refusal*: flag the prompt when its conformal p-value $(1 + \#\{k \leq K : S_k \geq S(\mathbf{e})\})/(K + 1)$ is at most β; this is the test of Proposition 1, with false-alarm rate at most β on safe prompts.
- *One-step blending* towards a feasible safe prompt (Proposition 8): closed form, two network evaluations.
- *Multi-step blending with back-off*: per-step blending of the velocities, with the bisection of Proposition 6 on the map metric.
- *Prompt projection* (Proposition 9): a second-order cone program over the simplex of the safe prompts and the request; for nonlinear models, a first-order step followed by a check, or a projected gradient method on $\mathbf{e}$ through the network.
- *Per-step guidance* (heuristic): at every step, project the current velocity onto $\{\mathbf{u} : \frac{1}{M} \sum_i \|\mathbf{u} - v(\mathbf{x}, t; \mathbf{e}_i)\| \leq r_t\}$. This is cheap, but it compares velocities at the same point rather than along the safe trajectories, so it carries no certificate without the Lipschitz arguments of Proposition 3.
- *Training-time enforcement*: fine-tune the conditional model with a penalty $\mathbb{E}_{\mathbf{e}}[(S(\mathbf{e}; \mathcal{D}) - \varepsilon)_+]$ over a distribution of prompts, so that the model maps every prompt into the region by itself. This is a data-driven counterpart of concept erasure (Gandikota et al. 2023), where the safe set rather than a list of forbidden concepts defines what is kept.

9.4 Calibration

- *Split conformal* (Definition 2): the procedure of this note.

- *Jackknife+ / cross-conformal* (Barber et al. 2021): every safe prompt serves both as anchor and as calibration point, at the price of a coverage of $1 - 2\beta$.
- *Mondrian*: calibration within clusters of prompts (Vovk et al. 2005).
- *Weighted conformal* (Barber et al. 2023): for a distribution of safe prompts that drifts over time.
- *Training-conditional (PAC)* (Vovk 2013): for a region deployed once and used for many queries.
- *Conformal risk control* (Angelopoulos et al. 2024): to choose the blending strength or the budget so that an expected loss that is monotone in the tuned parameter (e.g. an unsafe rate measured by a judge on a labelled calibration set, or a fidelity loss) is controlled, rather than a coverage.

9.5 Certificates in practice

The certificates of Theorems 1, 2 and 3 involve exact quantities; estimating them gives plug-in certificates (Remark 3). They become rigorous with:

- an upper confidence bound on each $d_\chi(\mathbf{e}, \mathbf{e}_i)^2$ computed on fresh probes (Hoeffding or empirical Bernstein when $\|\chi\|$ is bounded), combined over i by a union bound;
- a Clopper–Pearson upper bound on $\bar{P}_w(B^{+r})$, or on $\bar{P}_w(A)$, from the stored safe outputs;
- an upper confidence bound on the straightness defects $\zeta_{\mathbf{e}}$ in Theorem 3(iii).

For the divergences, the per-step terms of Corollary 1 are unbounded, and a rigorous bound needs a tail assumption or a robust estimator.

10 Experiment plan

The toy experiment of Section 7 tests the general theory with exact or linear fields. This section proposes experiments for the prompt-conditioned setting of Section 8 and for the design choices of Section 9, in increasing order of cost. Each experiment states what it tests and what would count as a failure.

10.1 Hypotheses

H1 (validity). For every cost and region, the false-alarm rate on held-out safe prompts is at most β up to sampling error (Proposition 1). Under a shift of the distribution of safe prompts it may exceed β, and weighted conformal prediction should repair it.

H2 (direction and sensitivity). At an equal false-alarm rate, the reverse divergence and the map metrics detect prompts with a small unsafe mass m that forward and fixed-reference costs miss. For map metrics the detection threshold scales like $m \approx ((M+1)\varepsilon/\Delta_{\mathrm{s}})^2$ (Proposition 7).

H3 (straightness). For straight (reflowed or one-step) models, $\zeta_{\mathbf{e}}$ is small, $d_0 \approx d_T$ (Lemma 8), and the certificate of Theorem 3 is non-vacuous at one network evaluation per probe. For 1-rectified models $\zeta_{\mathbf{e}}$ is larger and loosens the bound.

H4 (locality). On diverse safe sets, prompt-adaptive and order-∞ regions give tighter certificates and smaller fidelity losses than the uniform order-1 region, at the same false-alarm rate.

H5 (forcing). Forcing lowers the unsafe rate of unsafe prompts below the certificate, preserves prompt alignment better than refusal, and compares favourably with guidance-based and erasure-based baselines on the trade-off between unsafe rate and fidelity.

H6 (certificates under attack). Prompts optimised to maximise the unsafe rate subject to the score constraint never exceed the certificate; the gap measures its tightness.

10.2 Protocol common to all experiments

Prompt splits. Safe training prompts (M, defining the safe fields), safe calibration prompts (K), safe test prompts, unsafe test prompts (from a benchmark or constructed), and adversarial prompts (optimised against the filter, H6). The conformal statements are evaluated over repeated random splits of the safe prompts, as in Section 7.

Readouts.

1. The false-alarm rate on safe test prompts (H1).
2. The detection rate on unsafe test prompts at a fixed β, and the curve as β varies (H2).
3. The unsafe rate of the generations: exact membership in the unsafe set for the toys, an external judge otherwise (a held-out classifier, distinct from the χ used in the cost).

Table 6: Proposed experiments, in increasing order of cost.

experiment	model	safe and unsafe prompts	costs compared	hypotheses
E1 prompt-conditioned toy	linear conditional flow matching, $v = \Phi(\mathbf{x}, t)\Theta\mathbf{e}$, three couplings	four-blob rings ($u = 0$) vs. requests moving mass $m(u)$ to the centre	all of Table 5	H1–H3, H5
E2 straightness	class-conditional rectified flows on MNIST and CIFAR-10, with reflow and one-step distillation	class embeddings and their interpolations	d_T vs. d_0	H3
E3 forbidden class	same models	allowed classes and their interpolations vs. a forbidden class and interpolations towards it	d_T, d_χ, reverse KL	H2, H4, H5
E4 text-to-image	pretrained rectified-flow model, or a one-step reflowed model	curated captions vs. I2P and adversarial prompts	d_χ with an image encoder	H1, H2, H4, H5
E5 flow policies	conditional flow policy for 2-D navigation with obstacles	goals in the demonstrated region vs. goals behind obstacles or out of distribution	map metric on trajectories, reverse KL	H2, H5
E6 ablations and attacks	E1, E3, E5	as above, plus prompts optimised against the filter	all	H6

4. Fidelity on safe test prompts and on forced prompts: prompt alignment, e.g. the CLIP score (Radford et al. 2021); sample quality, e.g. FID (Heusel et al. 2017) and precision and recall (Kynkäänniemi et al. 2019); and the fraction of safe prompts whose generations the filter changes.
5. Certificate tightness: the bound of Theorem 1, 2 or 3 against the measured unsafe rate.
6. Cost: network evaluations per generated sample, and memory of the stored safe outputs.

Statistics. Means with bootstrap confidence intervals over prompts and seeds, and coverage over random splits.

10.3 Experiments

Table 6 summarises the experiments.

E1: prompt-conditioned toy. This is the cheapest test and extends the code of this folder. The model is a linear conditional flow-matching field $v(\mathbf{x}, t; \mathbf{e}) = \Phi(\mathbf{x}, t)\Theta\mathbf{e}$, exactly affine in the prompt, fitted by least squares on (prompt, sample) pairs. The prompts are $\mathbf{e} = (\cos\phi, \sin\phi, \varrho, u, 1)$ for the four-blob family of Section 7, where $u \in [0, 1]$ moves a mass $m(u)$ into the centre disk ($u = 0$ for safe prompts); the prompt vector can be enriched with random features of (ϕ, ϱ, u), which keeps the model affine in it. Three couplings vary the straightness: independent (1-rectified), one reflow step, and a minibatch optimal-transport coupling. *Readouts:* $\zeta_\mathbf{e}$ and $|d_T - d_0|$ (H3); coverage (H1); detection as a function of (m, Δ_s), moving the unsafe disk to vary Δ_s (H2); forcing by one-step blending and by prompt projection, which is exact here (H5); certificate against the measured unsafe rate. *Failure:* detection that does not follow the $\sqrt{m}\,\Delta_\mathrm{s}$ law, or a violated certificate, which would point to an error in Theorem 3 or in the estimation.

E2: straightness on images. Train class-conditional rectified flows on MNIST (LeCun et al. 1998) and CIFAR-10 (Krizhevsky 2009) with independent coupling, after one reflow step, and after one-step distillation. Prompts are class embeddings and their interpolations. *Readouts:* the distribution of $\zeta_\mathbf{e}$ over prompts; the correlation between d_0 and d_T; the tightness of $|d_T - d_0| \le \zeta_\mathbf{e} + \zeta_{\mathbf{e}'}$; the cost saved by d_0 (H3).

E3: a forbidden class. Use the same models. The safe prompts are the embeddings of the allowed classes and their interpolations; the unsafe prompts are the forbidden class and interpolations towards it with weight $u \in [0, 1]$. Compare d_T in pixel space, d_χ with χ the penultimate layer of a classifier trained on a separate split, and the reverse divergence with SDE sampling. *Readouts:* detection as a function of u (H2); uniform vs. prompt-adaptive vs. order-∞ regions (H4); the unsafe rate measured by an independent judge classifier, and FID and precision–recall on the allowed classes, before and after forcing (H5).

E4: text-to-image. Use a pretrained rectified-flow text-to-image model (Esser et al. 2024) or a one-step reflowed model (Liu et al. 2024). Safe prompts are captions of a curated dataset such as COCO (Lin et al. 2014). Unsafe prompts come from the I2P benchmark (Schramowski et al. 2023), plus adversarial prompts in the style of Ring-A-Bell (Tsai et al. 2024). The cost is d_χ with χ a CLIP image encoder on the decoded images. The safe outputs are stored on L probe noises, and the nearest safe prompts are retrieved in text-embedding space, both for the adaptive weights and for the forcing target $\mathbf{e}_c$. Baselines:

- negative prompts and the safety guidance of Schramowski et al. (2023);

- concept erasure (Gandikota et al. 2023);
- post-hoc filtering of the outputs.

Readouts: all six of Section 10.2, at scale. Screen generated content with automated judges and restrict access to it.

E5: flow policies. In a two-dimensional navigation task with obstacles, a conditional flow policy, in the spirit of diffusion policies (Chi et al. 2023), generates trajectories given a start and a goal, trained on demonstrations of a safe planner. Safe prompts are goals in the demonstrated region; test prompts include goals behind obstacles and out of distribution; the unsafe event is a collision. In low dimension the reverse divergence and the rare-event certificate of Theorem 2 are computable, so this is the natural place to compare them with the map metric on whole trajectories (χ returns the trajectory). *Readouts:* collision and success rates, and certificate tightness for rare collisions (H2, H5).

E6: ablations and attacks. On E1, E3 and E5, vary M, K, β, the number of probes L, plug-in vs. rigorous certificates, the weights (k, h), order-1 vs. order-∞, the costs of Table 5, the forcing variants, and straight vs. curved models. For H6, run a white-box attack: maximise the judge's unsafe score over prompt embeddings subject to $S(\mathbf{e}; \mathcal{D}) \leq \varepsilon$ by projected gradient on $\mathbf{e}$, and compare the unsafe rate it achieves with the certificate.

11 Discussion, limitations and next steps

What the note establishes. The order-1 machinery of the manuscript extends from weights to time-dependent vector fields without change, because its closed form only needs a pseudometric and its coverage only needs a fixed score. Viewing samplers as Gaussian-increment chains gives exact path divergences between fields, with time weights fixed by the sampler, and turns region membership into safety statements about the generated trajectories that hold for every event at once. For linear flow matching the whole construction reduces to the manuscript's Euclidean region after a whitening by the average historical Gram matrix. For prompt-conditioned models, where safe prompts induce safe fields, straight flows collapse time onto the slice $t = 0$, the distance between noise-to-output maps carries a Wasserstein certificate without Grönwall factor (Theorem 3), and forcing a prompt into the safe set has closed forms (Propositions 8 and 9).

What the toy experiment shows. Coverage holds at the nominal level. The forward/reverse gap predicted by Proposition 2 is clearly visible: convex forward costs admit fields with a few percent of unsafe mass, reverse costs do not. The certificates hold on every split, and the reverse-KL certificate is informative (0.031 with exact scores). Constrained fine-tuning improves on least squares in accuracy and in unsafe mass, but not on well-tuned shrinkage in this concentrated family.

Limitations. (i) The experiment is two-dimensional, with exact or linear fields and $M = 20$; nothing here tests neural score networks or high-dimensional data, where path divergences between models trained on different data are large and the certificates of Theorems 1–2 would be vacuous (Remark 4). The natural regime is a family of closely related safe fields, e.g. fine-tunes of one base model or diffusion policies for related tasks. (ii) Certificates are plug-in (Remark 3). (iii) The on-policy costs are non-convex; Algorithm 1 has no convergence guarantee and relies on the back-off for feasibility. (iv) Tasks are i.i.d.; historical fields that drift over time need weighted conformal prediction (Barber et al. 2023). (v) For deterministic samplers, the trajectory-wise certificate of Proposition 3 is vacuous here; the map certificate of Theorem 3 avoids the Grönwall factor but has not been tested. (vi) None of the prompt-conditioned results of Section 8 has been tested yet; Section 10 lists the experiments that would test them. (vii) Map metrics are only as semantic as the space chosen by χ, and every certificate is only as tight as the safe prompts are concentrated (Remark 4); for diverse prompt sets the local regions of Section 8.4 are needed.

Next steps. The experiments of Section 10, starting with the prompt-conditioned toy E1, which reuses the code of this folder. Beyond them: neural score or velocity networks trained with a primal–dual method and pathwise gradients through the sampler for the reverse cost; penalty plus constraint for few-shot accuracy; and rigorous Monte Carlo certificates (Section 9.5). The closest existing work calibrates the safety score of the outputs of a diffusion model rather than a region of fields (Hu et al. 2025), and Fan et al. (2023) regularise towards a single pretrained model with the per-step divergence of Corollary 1(i); we are not aware of work that calibrates a region around several historical fields.

A Reproduction

All scripts are in `code/` and write to `results/`; figures and tables are generated from `results/` only. Seeds are fixed, and every script is deterministic (the refactored fine-tuning worker reproduces the original run to $2 \cdot 10^{-13}$, and the rerun of pilot D reproduces its log exactly).

script	content	time
`part_a.py`	pilot A: exact-score chains, pairwise path divergences (forward and reverse), contaminated fields, $P^v(A)$; writes `part_a_cache.npz`	6.5 min
`part_a_analyze.py`	pilot A: coverage, exclusion and certificates over 200 splits; writes `part_a_splits.json`	10 s
`part_c_pre.py`	pilots C: ridge fits, contaminated fits, on-policy Grams, $P^v(A)$, coverage and exclusion over 100 splits; writes `part_c_state.npz`	18 min
`part_c_ft.py`	few-shot fine-tuning, one worker per range of test tasks (8 workers in parallel); writes `part_c_ft_*.json`	20 min
`part_d.py`	pilot D: SDE chains of the linear fits, KL regions, certificates; writes `part_d_cache.npz`, `part_d_splits.json`	11 min
`example_fits.py`, `time_profiles.py`	data for Figures 1, 4 and 6–8	2 min
`make_figures.py`, `make_tables.py`	all figures, tables and the numbers quoted in the text	1 min

The correspondence between the notation of the note and the code is: J is `K` in the code (number of time steps); $\boldsymbol{\theta}$ is the matrix $\mathtt{W} \in \mathbb{R}^{2\times 262}$ with $\boldsymbol{\theta} = \mathrm{vec}(\mathtt{W}^\top)$; $G_\rho = I_2 \otimes G_\psi$ and $\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_G = \|(\mathtt{W} - \mathtt{W}')C_\psi\|_F$ with $G_\psi = C_\psi C_\psi^\top$; the fixed-Gram and on-policy regions are `ref` and `on` in pilot C, and the forward and reverse KL regions are `fwd` and `rev` in pilot D (`F`, `R`, `K` in pilot A).

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